THE MATHEMATICS OF LATTICE-BASED CRYPTOGRAPHY

1. Introduction

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NIST's PQC standards

- In August 2024, the US government's National Institute of Standards and Technology (NIST) published a suite of standards for key encapsulation and digital signatures schemes that are resistant to attacks by quantum computers.
- These quantum-safe schemes are intended to replace their RSA and ECC counterparts, which we know can be completely broken by attacks that run on quantum computers.
- Even though the timeline for building cryptographically-relevant quantum computers remains uncertain, the effort to replace ECC and RSA with quantum-safe algorithms is accelerating.

Kyber and Dilithium

- The quantum-safe schemes that are expected to see the most deployment in the coming years are **Kyber** and **Dilithium**.
- The security of Kyber is based on the hardness of the **Decisional-Module-Learning With Errors** (D-MLWE) problem, which in turn is related to the hardness of the **Module Learning With Errors** (MLWE) problem.
- The security of Dilithium is based on the hardness of D-MLWE and also the **Module Short Integer Solutions** (MSIS) problem.

Lattice-based cryptosystems

- In my short course "Kyber and Dilithium", I gave detailed descriptions of Kyber (ML-KEM) and Dilithium (ML-DSA) as standardized in the NIST standards FIPS 203 and FIPS 204.
- I didn't explain the connection between Kyber/Dilithium and lattices.
- The purpose of this course is to fill this gap, and explain why the MSIS and MLWE problems can be viewed as computational problems with lattices.
- This will justify calling Kyber and Dilithium **lattice-based cryptosystems**.

Course outline

1. Introduction
2. Short Integer Solutions (SIS) problem
3. Learning With Errors (LWE) problem
4. Lattices
5. SIS/LWE and lattices
6. Ring-SIS and Ring-LWE
7. Module-SIS and Module-LWE

Course material

- The course webpage is cryptography101.ca, where you can find links to the Youtube **videos** and the lecture **slides**.
- If you are just getting started in post-quantum cryptography, you should first watch the following videos from my "Kyber and Dilithium" course:
  - **V1a**: Post-quantum cryptography
  - **V1b**: Mathematical prerequisites
- My other online courses are:
  - Kyber and Dilithium
  - Cryptography 101: Building Blocks
  - Cryptography 101: Deployments (coming in 2025)
  - Error-Correcting Codes

Please subscribe to my YouTube channel and recommend my courses to your colleagues and fellow students.

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2. Short Integer Solutions (SIS) Problem

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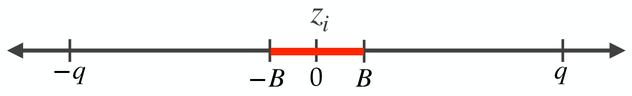
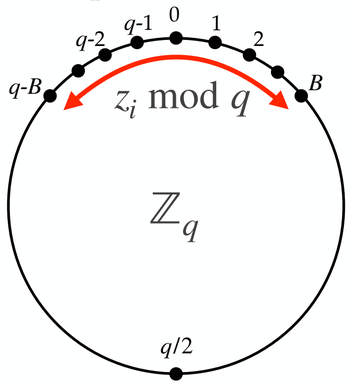
Outline

1. SIS definition
2. Collision-resistant hash function
3. ISIS definition
4. SIS and ISIS are equivalent
5. Normal-form ISIS

SIS definition

- SIS was introduced by Ajtai in 1996.
- **Definition**. (*Homogeneous*) *Short Integer Solutions problem*: SIS(n, m, q, B)
  Given $A \in_R \mathbb{Z}_q^{n\times m}$, find $z \in \mathbb{Z}^m$ such that $Az = 0 \pmod q$, where $z \neq 0$ and $z \in [-B, B]^m$ (and $B \ll q/2$).



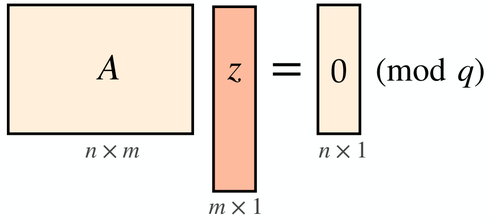


- **Notation**:
  1. $\mathbb{Z}_q = \{0,1,2,\ldots,q-1\}$.
  2. $x \in_R S$ means that x is selected uniformly (and independently) at random from S.
  3. All vectors are column vectors.

Existence of an SIS solution

SIS: Given $A \in_R \mathbb{Z}_q^{n\times m}$, find $z \in \mathbb{Z}^m$ such that $Az = 0 \pmod q$, where $z \neq 0$ and $z \in [-B, B]^m$ (and $B \ll q/2$).

1. If $n \geq m$, then one expects that $Az = 0 \pmod q$ has a unique solution $z = 0$, so no SIS solution exists. Henceforth, we'll assume that $n < m$.
2. If $(B+1)^m > q^n$, then by the pigeonhole principle there must exist $z_1, z_2 \in [-B/2, B/2]^m$ such that $z_1 \neq z_2$ and $Az_1 = Az_2 \pmod q$.
   Then $z = z_1 - z_2$ is an SIS solution.
3. So, we'll henceforth assume that $(B+1)^m > q^n$, i.e., $m > (n \log q)/\log(B+1)$, whereby an SIS solution is guaranteed to exist.
4. The SIS solution is not unique.
   Indeed, if z is an SIS solution, then so is $-z \bmod q$.



SIS example

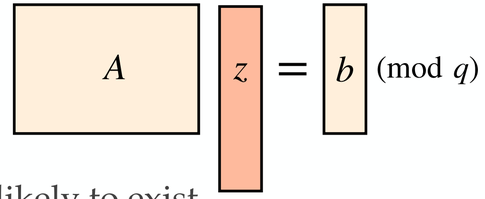
- Let $n = 3$, $m = 5$, $q = 13$, and $B = 3$.
- **SIS instance:**
  $$A = \begin{bmatrix} 1 & 0 & 7 & 12 & 4 \\ 2 & 11 & 3 & 6 & 12 \\ 9 & 8 & 10 & 5 & 1 \end{bmatrix}.$$
- We need to find nonzero $z = (z_1, z_2, z_3, z_4, z_5) \in [-3,3]^5$ with $Az = 0 \pmod{13}$.
- Equivalently, we have $z_i \bmod 13 \in \{0,1,2,3,10,11,12\}$.
- Performing Gaussian elimination on A yields the reduced matrix $A' = \begin{bmatrix} 1 & 0 & 0 & 5 & 10 \\ 0 & 1 & 0 & 10 & 12 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$.
- Thus the complete solution to $Az = 0 \pmod{13}$ is $z_1 = 8z_4 + 3z_5$, $z_2 = 3z_4 + z_5$, $z_3 = 12z_4 + 12z_5$, $z_4 \in \mathbb{Z}_{13}$, $z_5 \in \mathbb{Z}_{13}$.
- Among the $13^2 = 169$ solutions $z \in \mathbb{Z}_{13}^5$, are **six SIS solutions:**
  $z = \pm(3,1,-1,0,1)$, $z = \pm(1,0,-2,-1,3)$, and $z = \pm(2,1,1,1,-2)$.

SIS application: Collision-resistant hash function

- **Hash function definition**:
  - Select $A \in_R \mathbb{Z}_q^{n\times m}$, where $m > n \log q$.
  - Define $H_A : \{0,1\}^m \longrightarrow \mathbb{Z}_q^n$ by $H_A(z) = Az \pmod q$.
- **Notes**
  1. **Compression**. Since $m > n \log q$, we have $2^m > q^n$.
     Thus, H_A is indeed a compression function.
  2. **Collision resistance**. Suppose that one can efficiently find $z_1, z_2 \in \{0,1\}^m$ with $z_1 \neq z_2$ and $H_A(z_1) = H_A(z_2)$. Then $Az_1 = Az_2 \pmod q$, whence $Az = 0 \pmod q$ where $z = z_1 - z_2$. Since $z \neq 0$ and $z \in [-1,1]^m$, z is an SIS solution (with $B = 1$) which has been efficiently found. $\square$

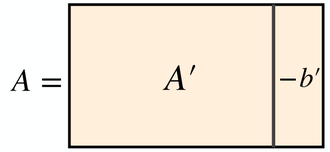
Inhomogeneous SIS (ISIS)

- **Definition**. *Inhomogeneous Short Integer Solutions problem*: ISIS(n, m, q, B)
  Given $A \in_R \mathbb{Z}_q^{n\times m}$ and $b \in_R \mathbb{Z}_q^n$, find $z \in \mathbb{Z}^m$ such that $Az = b \pmod q$ and $z \in [-B, B]^m$.



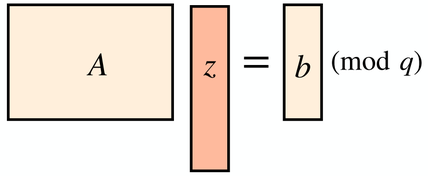
- **Notes**
  1. We'll assume that $n < m$.
  2. If $(2B+1)^m \gg q^n$, then an ISIS solution is likely to exist.
  3. So, we'll henceforth assume that $(2B+1)^m \gg q^n$, i.e., $m \gg (n \log q)/\log(2B+1)$.

SIS and ISIS are equivalent (1)



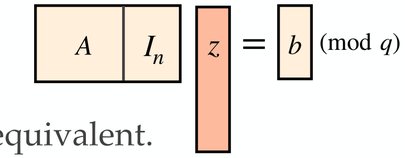
- **Claim 1**. SIS $\leq$ ISIS.
- **Proof**. Let A be an SIS instance.
  Write $A = [A' \mid -b']$ where $A' \in \mathbb{Z}_q^{n\times(m-1)}$ and $b' \in \mathbb{Z}_q^n$.
  Determine an ISIS solution z' to the ISIS instance (A', b'), so $A'z' = b' \pmod q$ and $z' \in [-B, B]^{m-1}$.
  Then $z = \begin{bmatrix} z' \\ 1 \end{bmatrix} \in \mathbb{Z}^m$ satisfies $Az = 0 \pmod q$, $z \neq 0$, and $z \in [-B, B]^m$.
  Thus, z is an SIS solution that we have efficiently found. $\square$

SIS and ISIS are equivalent (2)



- **Claim 2**. ISIS $\leq$ SIS.
- **Proof**. Let (A, b) be an ISIS instance.
  Select $j \in_R [1, m+1]$ and $c \in_R [-B, B]$ with $c \neq 0$.
  Let A' be the $n \times (m+1)$ matrix obtained by inserting $-c^{-1}b \bmod q$ as a new jth column in A.
  Determine an SIS solution $z' \in [-B, B]^{m+1}$ to $A'z' = 0 \pmod q$.
  If indeed the jth entry in z' is c, then $Az = b \pmod q$, where $z \in [-B, B]^m$ is obtained from z' by deleting its jth entry.
  Thus, z is an ISIS solution that we have efficiently found. $\square$

Normal-form ISIS (nf-ISIS)



- **Definition**. *Normal-form ISIS problem*: nf-ISIS(n, m, q, B)
  Given $A \in_R \mathbb{Z}_q^{n\times m}$ and $b \in_R \mathbb{Z}_q^n$, find $z \in \mathbb{Z}^{m+n}$ such that $[A \mid I_n]z = b \pmod q$ and $z \in [-B, B]^{m+n}$.
- **Claim**. nf-ISIS(n, m, q, B) and ISIS$(n, m+n, q, B)$ are equivalent.
- **Proof**. (nf-ISIS $\leq$ ISIS) Given a nf-ISIS instance (A, b), select $C \in_R \mathbb{Z}_q^{n\times n}$; C is invertible with probability roughly $(q-1)/q$. Then $([CA \mid C], Cb)$ is an ISIS instance with the same solution space as the nf-ISIS instance.
- (ISIS $\leq$ nf-ISIS) Given an ISIS instance (A, b), write $A = [A' \mid C]$; note that C is invertible with probability roughly $(q-1)/q$. Then $([C^{-1}A' \mid I_n], C^{-1}b)$ is a nf-ISIS instance with the same solution space as the ISIS instance. $\square$

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3. Learning With Errors (LWE) Problem

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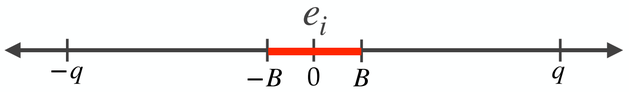
Outline

1. LWE definition
2. LWE parameters
3. Decisional LWE
4. Short-secret LWE (ss-LWE)
5. Application: Lindner-Peikert public-key encryption scheme

LWE definition

- **Notation**:
  - $\mathbb{Z}_q = \{0,1,2,\ldots,q-1\}$.
  - $x \in_R S$ means that x is selected uniformly (and independently) at random from S.
  - All vectors are column vectors.
- LWE was introduced by Regev in 2005.
- **Definition**. *Learning With Errors problem*: LWE(m,n,q,B)
  Let $s \in_R \mathbb{Z}_q^n$ and $e \in_R [-B,B]^m$ where $B \ll q/2$.
  Given $A \in_R \mathbb{Z}_q^{m\times n}$ and $b = As + e \pmod q \in \mathbb{Z}_q^m$, find s.



LWE example

- Let $m = 5$, $n = 3$, $q = 31$, and $B = 2$.
- **LWE instance:**

$$A = \begin{bmatrix} 11 & 3 & 27 \\ 12 & 21 & 7 \\ 6 & 23 & 30 \\ 5 & 6 & 2 \\ 21 & 0 & 14 \end{bmatrix}, \quad b = \begin{bmatrix} 25 \\ 25 \\ 12 \\ 29 \\ 17 \end{bmatrix}.$$

- **LWE challenge**: We need to find $s \in \mathbb{Z}_{31}^3$ and $e \in [-2,2]^5$ with $As + e = b \pmod{31}$.
- In fact, there are *three* LWE solutions:
  $s = (2,11,7)^T$, $e = (-2,0,2,1,1)^T$,
  $s = (27,13,16)^T$, $e = (1,-2,1,1,1)^T$,
  $s = (30,9,5^T)$, $e = (-2,-1,2,1,-1)^T$.
- In general, one cannot guarantee that there is a unique LWE solution.
- So, the LWE parameters must be carefully selected so that the probability that an LWE instance has more than one solution is negligibly small.

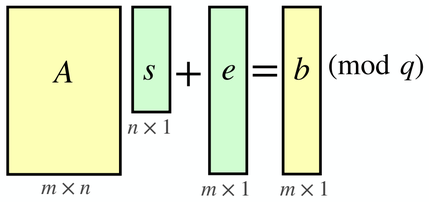
LWE parameter B

1. If $B = 0$ (so $e = 0$), then $As = b \pmod q$ can be solved efficiently.

2. If $B = (q-1)/2$, then finding s is information-theoretically impossible. So, we'll henceforth assume that $B < q/4$.

3. (Arora-Ge) If B is asymptotically smaller than $\sqrt{n}$, then LWE can be solved in subexponential time for sufficiently large $m \gg n$.

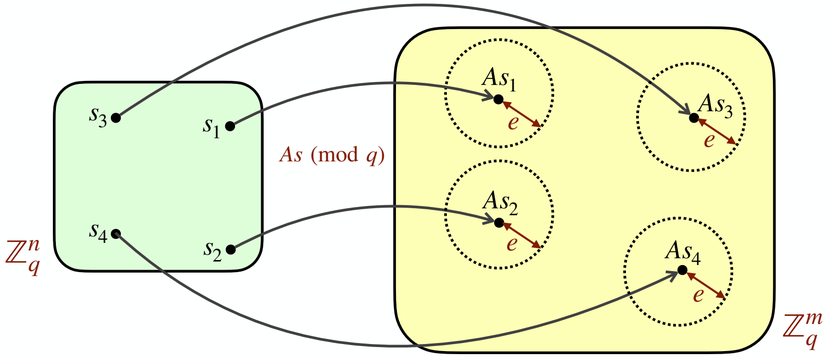
LWE: Let $s \in_R \mathbb{Z}_q^n$ and $e \in_R [-B, B]^m$ where $B \ll q/2$. Given $A \in_R \mathbb{Z}_q^{m \times n}$ and $b = As + e \pmod q \in \mathbb{Z}_q^m$, find s.

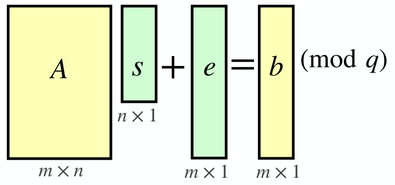


LWE parameters m and n

- If $m \gg n$, then one expects that there is a unique LWE solution (s, e).
- Henceforth, we'll assume that $m \gg n$.

LWE: Let $s \in_R \mathbb{Z}_q^n$ and $e \in_R [-B, B]^m$ where $B \ll q/2$. Given $A \in_R \mathbb{Z}_q^{m \times n}$ and $b = As + e \pmod q \in \mathbb{Z}_q^m$, find s.

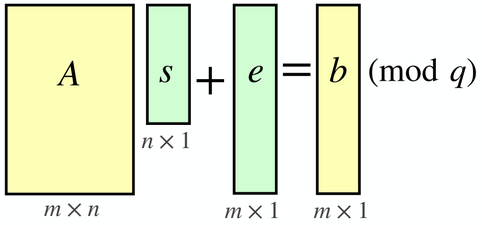


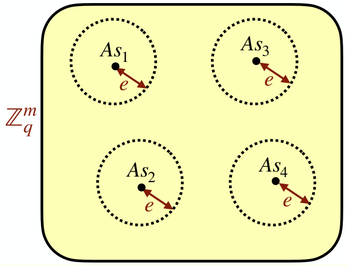


Uniqueness of the LWE solution is only guaranteed if no two of the q^n spheres centred at the vectors $As \pmod q$ overlap.

Decisional LWE (DLWE)

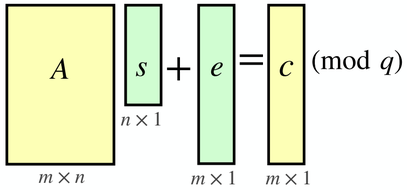
Definition. *Decisional LWE problem*: DLWE(m, n, q, B)
Let $A \in_R \mathbb{Z}_q^{m \times n}$, $s \in_R \mathbb{Z}_q^n$, $e \in_R [-B, B]^m$ where $B \ll q/2$, and $b = As + e$.
Let $r \in_R \mathbb{Z}_q^m$.
Let $c = b$ with probability $1/2$, and $c = r$ with probability $1/2$.
Given (A, c), the problem is to decide (with success probability significantly greater than $1/2$) whether $c = b$ or $c = r$.





DLWE and LWE are equivalent (1)

- **Claim**. DLWE and LWE are equivalent.
- **Claim 1**. DLWE $\leq$ LWE.
- **Proof**. Let (A, c) be a DLWE instance.
  Now, if $c = b$, then (A, c) is an LWE instance and so one expects that $As + e = c \pmod q$ has a unique LWE solution (s, e) (with $e \in [-B, B]^m$).
  And, if $c = r$, then one expects that $As + e = c$ does not have an LWE solution.
  So, the LWE solver is run with input (A, c). If a valid LWE solution is returned, then one concludes that $c = b$. If the LWE solver terminates without a valid LWE solution, or fails to terminate, then one concludes that $c = r$. $\square$

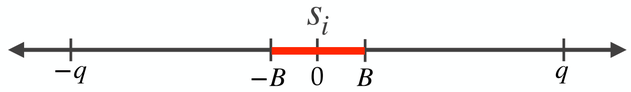


DLWE and LWE are equivalent (2)

- **Claim 2** LWE $\leq$ DLWE.
- **Proof**. Let (A, b) be an LWE instance (where $b = As + e$). We'll use a DLWE solver to test our guesses for the coordinates of s, one at a time, beginning with s_1.
  Let $d \in \mathbb{Z}_q$. Here's how we test whether $s_1 = d$.
  Select $\Delta \in_R \mathbb{Z}_q^m$. Let A' be the matrix obtained by adding Δ to the first column of A, and let $b' = b + d\Delta$.
  Now, if $s_1 = d$, then $b' = A's + e$, so (A', b') is a valid LWE instance.
  On the other hand, if $s_1 \neq d$, then $b' = A's + e + (d - s_1)\Delta$. Since $d - s_1$ is nonzero, and Δ in uniformly random and independent of A', s and e, it follows that b' is uniformly random and independent of A'.
  Thus, the DLWE solver with input (A', b') will inform us whether or not $s_1 = d$. $\square$

Short-Secret LWE (ss-LWE)

- **Definition**. *Short-secret LWE problem*: ss-LWE(m, n, q, B)
  Let $s \in_R [-B, B]^n$ and $e \in_R [-B, B]^m$ where $B \ll q/2$.
  Given $A \in_R \mathbb{Z}_q^{m \times n}$ and $b = As + e \pmod q \in \mathbb{Z}_q^m$, find s.



- **Claim**. LWE and ss-LWE are equivalent.
  More precisely, ss-LWE$(m, n, q, B) \leq$ LWE(m, n, q, B) and LWE$(m, n, q, B) \leq$ ss-LWE$(m - n, n, q, B)$.
- **Exercise**. ss-LWE and ss-DLWE are equivalent.

ss-LWE and LWE are equivalent (1)

- **Claim 1**. ss-LWE$(m, n, q, B) \leq$ LWE(m, n, q, B).
- **Proof**. Let (A, b) be an ss-LWE(m, n, q, B) instance, where $b = As + e \pmod q$ and $s \in_R [-B, B]^n$ and $e \in_R [-B, B]^m$.
  Select $d \in_R \mathbb{Z}_q^n$ and let $b' = b + Ad = (As + e) + Ad = A(s + d) + e$.
  Then (A, b') is an LWE(m, n, q, B) instance.
  The solution (s', e) to this LWE instance, immediately gives the solution $(s' - d, e)$ to the ss-LWE instance. $\square$

ss-LWE and LWE are equivalent (2)

- **Claim 2**. LWE$(m, n, q, B) \leq$ ss-LWE$(m - n, n, q, B)$.
- **Proof**. Let (A, b) be an LWE(m, n, q, B) instance, so $b = As + e$.
  Let $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$, $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$, and $e = \begin{bmatrix} e_1 \\ e_2 \end{bmatrix}$ where $A_1, A_2, b_1, b_2, e_1, e_2$ have dimensions $n \times n$, $(m - n) \times n$, $n \times 1$, $(m - n) \times 1$, $n \times 1$, and $(m - n) \times 1$, respectively.
  Let $A' = -A_2 A_1^{-1} \in \mathbb{Z}_q^{(m-n) \times n}$ and $b' = A'b_1 + b_2 \in \mathbb{Z}_q^{m-n}$.
  Now, $b' = A'b_1 + b_2 = (-A_2 A_1^{-1})(A_1 s + e_1) + (A_2 s + e_2)$
  $= -A_2 s - A_2 A_1^{-1} e_1 + A_2 s + e_2 = A'e_1 + e_2$.
  Thus, (A', b') is an ss-LWE$(m - n, n, q, B)$ instance.
  A solution to the ss-LWE instance immediately gives a solution to the LWE instance. $\square$

PKE: Key generation

[Lindner-Peikert 2011]

Key generation: Alice does:

1. Select $s \in_R [-B, B]^n$ and $e \in_R [-B, B]^n$.
2. Select $A \in_R \mathbb{Z}_q^{n \times n}$.
3. Compute $b = As + e \pmod q$.
4. Alice's **public key** is (A, b); her **private key** is s.

- Determining any information about s from (A, b) is an instance of **ss-DLWE**(n, n, q, B).

PKE: Encryption and decryption

Encryption: To encrypt a message $m \in \{0,1\}$ for Alice, Bob does:

1. Obtain an authentic copy of Alice's encryption key (A, b).
2. Select $r, z \in_R [-B, B]^n$ and $z' \in_R [-B, B]$.
3. Compute $c_1 = A^T r + z$ and $c_2 = b^T r + z' + m\lceil q/2 \rfloor$.
4. Output $c = (c_1, c_2)$.

Note: $c \in \mathbb{Z}_q^n \times \mathbb{Z}_q$.

Decryption: To decrypt $c = (c_1, c_2)$, Alice does:

1. Output $m = \text{Round}_q(c_2 - s^T c_1)$.

Note: Alice uses her private key s.

Round$_q$: For $x \in [0, q-1]$, define

$$x \text{ mods } q = \begin{cases} x & \text{if } x \le (q-1)/2, \\ x - q & \text{if } x > (q-1)/2. \end{cases}$$

Then

$$\text{Round}_q(x) = \begin{cases} 0, & \text{if } -q/4 < x \text{ mods } q < q/4, \\ 1, & \text{otherwise}. \end{cases}$$

Toy example: PKE (1)

- **Domain parameters**: $n = 3,\ q = 229,\ B = 2$.
- **Key generation:** Alice selects:

$$A = \begin{bmatrix} 101 & 173 & 27 \\ 192 & 121 & 7 \\ 116 & 223 & 30 \end{bmatrix},\ s = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix},\ e = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix},\ \text{and computes}$$

$$b = As + e \pmod{229} = \begin{bmatrix} 112 \\ 147 \\ 17 \end{bmatrix}.$$

Alice's **encryption key** is (A, b); her **decryption key** is s.

Toy example: PKE (2)

- **Encryption**: To encrypt the plaintext bit $m = 1$, Bob selects

$$r = \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix},\ z = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix},\ z' = -2, \text{ and computes}$$

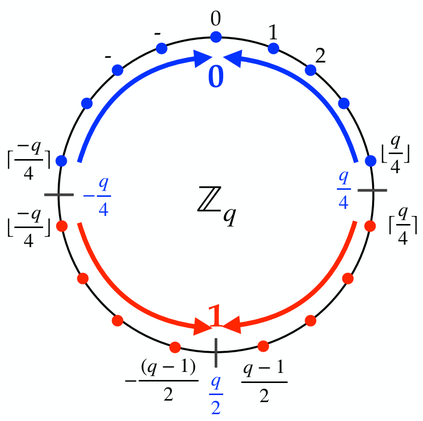
$$c_1 = A^T r + z \pmod{229} = \begin{bmatrix} 160 \\ 111 \\ 37 \end{bmatrix} \text{ and } c_2 = b^T r + z' + 115m \pmod{229} = 26.$$

The **ciphertext** is $c = (c_1, c_2)$.

- **Decryption**: To decrypt $c = (c_1, c_2)$, Alice uses her decryption key s to compute $c_2 - s^T c_1 \pmod{229} = 120$.
Now, $120 \text{ mods } 229 = -109$, and $\text{Round}_{229}(-109) = 1$.
Thus, Alice recovers the plaintext $m = 1$.

PKE: Decryption works

- **Question**: Does decryption work?
  i.e., does $m = \text{Round}_q(c_2 - s^T c_1)$?
- We have $c_2 - s^T c_1 = (b^T r + z' + m\lceil q/2 \rfloor) - s^T(A^T r + z)$
  $= (s^T A^T + e^T)r + z' + m\lceil q/2 \rfloor - s^T(A^T r + z)$
  $= e^T r - s^T z + z' + m\lceil q/2 \rfloor$.
- So, decryption works iff
  $|e^T r - s^T z + z' \bmod q| < q/4$.
- Now, suppose that $B \le \sqrt{q/(4(2n+1))}$.
- Then $|e^T r - s^T z + z' \bmod q| \le nB^2 + nB^2 + B \le \dfrac{2nq}{4(2n+1)} + \sqrt{\dfrac{q}{4(2n+1)}}$
  $= \dfrac{nq}{2(2n+1)} + \sqrt{\dfrac{q}{4(2n+1)}} < \dfrac{q}{4}$,
  so decryption works. □



PKE: Security

- **Claim**: The Lindner-Peikert PKE is indistinguishable against chosen-plaintext attack assuming that ss-DLWE is hard.
- **Proof**: The encryption operation can be written as: $\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} A^T \\ b^T \end{bmatrix} r + \begin{bmatrix} z \\ z' \end{bmatrix} + \begin{bmatrix} 0 \\ \lceil \frac{q}{2} \rfloor m \end{bmatrix}$.

  By the ss-DLWE assumption, $\begin{bmatrix} A^T \\ b^T \end{bmatrix}$ is indistinguishable from random.

  Again by the ss-DLWE assumption, $\begin{bmatrix} A^T \\ b^T \end{bmatrix} r + \begin{bmatrix} z \\ z' \end{bmatrix} = \begin{bmatrix} A^T r + z \\ b^T r + z' \end{bmatrix}$ is indistinguishable from random.

  Thus, from the adversary's perspective, c_2 appears to be the sum of the random element $b^T r + z' \in \mathbb{Z}_q$ and the plaintext $\lceil \frac{q}{2} \rfloor m$, so the adversary can learn nothing about m. □

THE MATHEMATICS OF LATTICE-BASED CRYPTOGRAPHY

4. Lattices

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Outline

1. Definition of a lattice
2. Characterization of the bases of a lattice
3. Successive minima
4. LLL lattice basis reduction algorithm
5. SVP
6. SIVP

Lattice definition

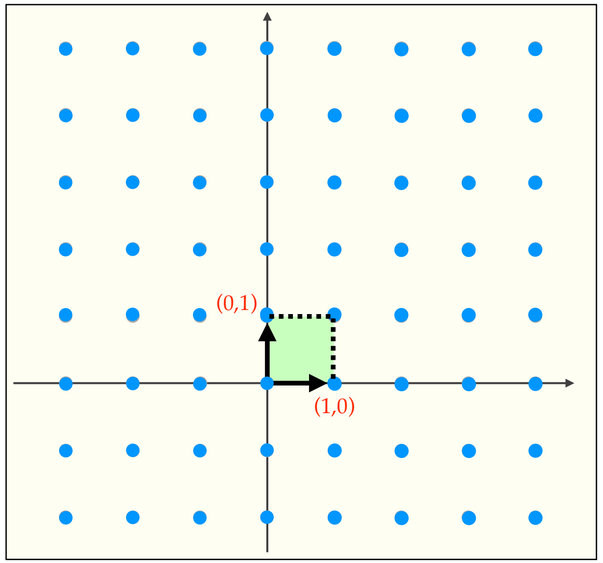
- **Definition**. A *lattice* L in $\mathbb{R}^n$ is the set of all <u>integer</u> linear combinations of m linearly independent vectors $B = \{v_1, v_2, \ldots, v_m\}$ in $\mathbb{R}^n$ (and where $m \le n$). The set B is called a *basis* of L, and we write $L = L(B)$. The *dimension* of L is n, and the *rank* of L is m.
- **Notes**:
  1. We will henceforth assume that the basis vectors $v_1, v_2, \ldots, v_m$ are in $\mathbb{Z}^n$.
  2. Thus, $L = \{x_1v_1 + x_2v_2 + \cdots + x_mv_m \ : \ x_1, x_2, \ldots, x_m \in \mathbb{Z}\} \subseteq \mathbb{Z}^n$.
     L is called an *integer lattice*.
  3. Let B be the $n \times m$ matrix whose columns are the basis vectors $v_1, \ldots, v_m$,
     so $B = \begin{bmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_m \\ | & | & \cdots & | \end{bmatrix}$. Then $L = \{Bx : x \in \mathbb{Z}^m\}$.

Full-rank lattices

- **Definition**. A *full-rank lattice* L in $\mathbb{R}^n$ is a lattice in $\mathbb{R}^n$ of rank n.
- **Definition**. Let L and L' be lattices in $\mathbb{R}^n$.
  Then L' is a *sublattice* of L if $L' \subseteq L$.
- Henceforth, unless otherwise stated, all lattices and sublattices will be full-rank (and integer).
- Note that a basis $B = \{v_1, v_2, \ldots, v_n\}$ for a full-rank lattice in $\mathbb{R}^n$ is also a basis for the vector space $\mathbb{R}^n$.

Lattice: Example 1

- Let $n = 2$ and $B_1 = \{(1,0), (0,1)\}$.
- Then $L_1 = L(B_1) = \{B_1x : x \in \mathbb{Z}^2\}$,
  where $B_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- Thus, $L_1 = \mathbb{Z}^n$.
- **Fundamental parallelepiped**:
  $P(B_1) = \{a_1(1,0) + a_2(0,1) : a_1, a_2 \in [0,1)\}$.

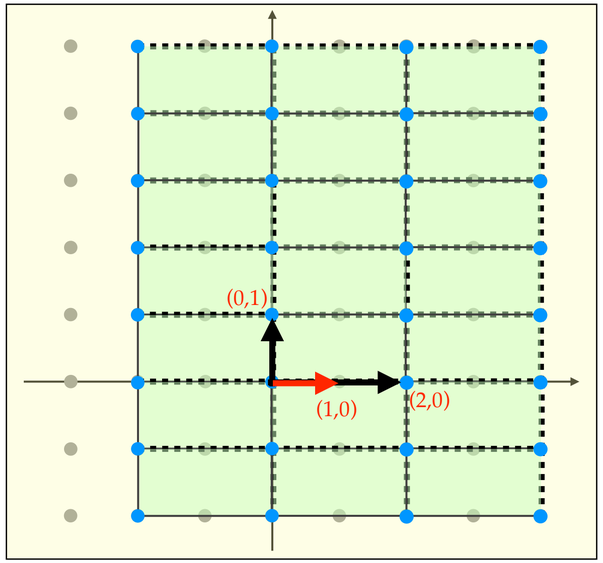


Fundamental parallelepiped

- **Definition**. Let $L = L(B)$ be a lattice in $\mathbb{R}^n$, where $B = \{v_1, v_2, \ldots, v_n\}$.
  The *fundamental parallelepiped* of L is
  $P(B) = \{a_1v_1 + a_2v_2 + \cdots + a_nv_n : a_i \in [0,1)\}$.
- **Notes**:
  1. Equivalently, $P(B) = \{Bx : x \in [0,1)^n\}$.
  2. $P(B)$ can be used to partition $\mathbb{R}^n$ into non-overlapping regions (called parallelepipeds). The "corners" of these parallelepipeds are the elements of the lattice $L(B)$.

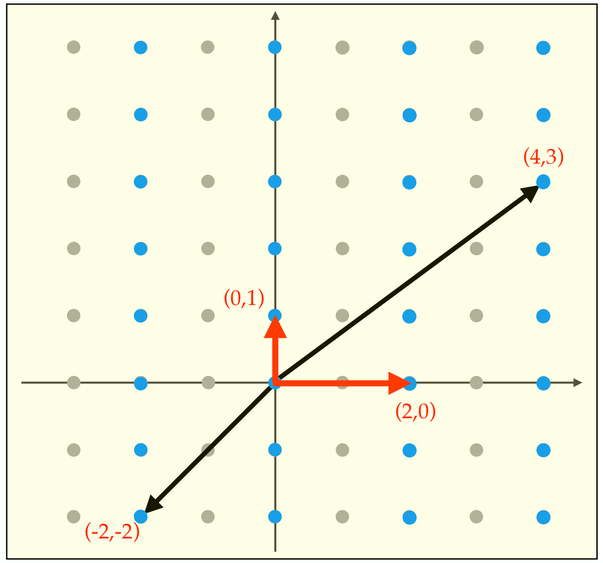
Lattice: Example 2

- Let $n = 2$ and $B_2 = \{(2,0), (0,1)\}$.
- Then
  $L_2 = L(B_2) = \{B_2 x : x \in \mathbb{Z}^2\}$,
  where $B_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$.
- **Notes**:
  1. L_2 a *sublattice* of L_1.
  2. $L_2 \neq L_1$ since $(1,0) \in L_1$, but $(1,0) = \frac{1}{2} \cdot (2,0) + 0 \cdot (0,1) \notin L_2$.



Lattice: Example 3

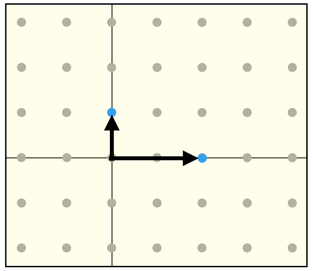
- Let $n = 2$ and $B_3 = \{(-2,-2), (4,3)\}$.
- Then $L_3 = L(B_3) = \{B_3 x : x \in \mathbb{Z}^2\}$, where $B_3 = \begin{bmatrix} -2 & 4 \\ -2 & 3 \end{bmatrix}$.
- **Notes**:
  1. $L_2 \subseteq L_3$ since
     $(2,0) = 3 \cdot (-2,-2) + 2 \cdot (4,3)$ and
     $(0,1) = -2 \cdot (-2,-2) - 1 \cdot (4,3)$.
  2. $L_3 \subseteq L_2$ since
     $(-2,-2) = -1 \cdot (2,0) - 2 \cdot (0,1)$ and
     $(4,3) = 2 \cdot (2,0) + 3 \cdot (0,1)$.
  3. Thus $L_3 = L_2$.



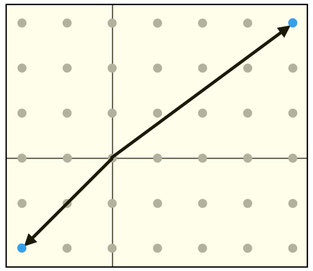
One basis is "nicer" than the other

- $L_2 = L(\{(2,0), (0,1)\}$ and
  $L_3 = L(\{(-2,-2), (4,3)\})$ are the same lattice, but described using different bases.
- The basis $B_2 = \{(2,0), (0,1)\}$ is "nicer" than the basis $B_3 = \{(-2,-2), (4,3)\}$ since the vectors in B_2 are "shorter" and "orthogonal" to each other.
- The *length* of a vector
  $a = (a_1, a_2, \ldots, a_n) \in \mathbb{R}^n$ is
  $\|a\|_2 = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$ (also called the *Euclidean length* or ℓ_2- *norm*).

B_2

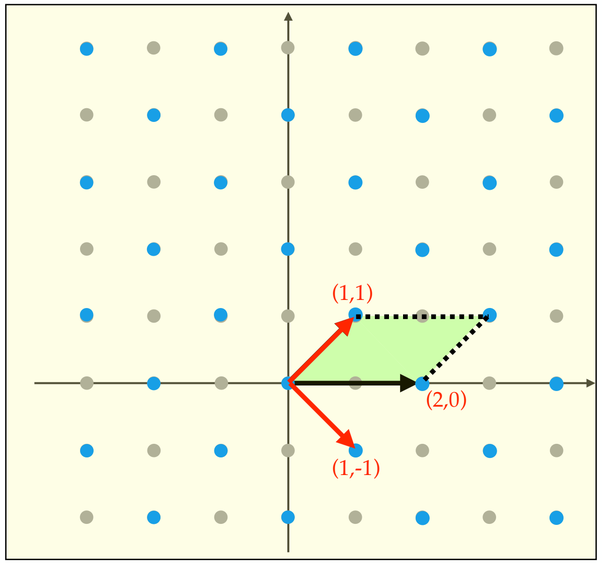


B_3



Lattice: Example 4

- Let $n = 2$ and $B_4 = \{(2,0), (1,1)\}$.
- Then $L_4 = L(B_4) = \{B_4 x : x \in \mathbb{Z}^2\}$,
  where $B_4 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$.
- <u>Exercise</u>: Prove that $L_4 \neq L_1$ and $L_4 \neq L_2$.
- <u>Exercise</u>: Prove that $\{(1,-1), (1,1)\}$ is another (nicer) basis for L_4.



A lattice has infinitely many bases

- **Theorem** (*characterization of lattice bases*) Let $L = L(B_1)$ be an n-dimensional (integer) lattice. Then an $n \times n$ integer matrix B_2 is also a basis for L if and only if $B_1 = B_2 U$, where U is an $n \times n$ matrix (the change-of-basis matrix) with integer entries and with $\det(U) = \pm 1$. (Such a matrix U is called *unimodular*.)
- **Example**. $B_2 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $B_3 = \begin{bmatrix} -2 & 4 \\ -2 & 3 \end{bmatrix}$ are bases for the same lattice since $\underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}}_{B_2} = \underbrace{\begin{bmatrix} -2 & 4 \\ -2 & 3 \end{bmatrix}}_{B_3} \cdot \underbrace{\begin{bmatrix} 3 & -2 \\ 2 & -1 \end{bmatrix}}_{U}$ where U is a unimodular matrix.

Proof of the characterization of lattice bases

- **Proof**. ($\Longrightarrow$) Suppose that B_1 and B_2 are both bases for $L \subseteq \mathbb{R}^n$.
  Since B_1 is a basis for L, and since the vectors in B_2 are in L, we can write $B_2 = B_1 U$ for some invertible matrix $U \in \mathbb{Z}^{n \times n}$.
  Similarly, we can write $B_1 = B_2 V$ for some invertible matrix $V \in \mathbb{Z}^{n \times n}$.
  Now, $B_1 = B_2 V = (B_1 U)V = B_1(UV)$.
  Since B_1 is invertible, we have $UV = I_n$.
  Thus, $\det(U)\det(V) = 1$, and hence $\det(U) = \pm 1$ and $\det(V) = \pm 1$.

  ($\Longleftarrow$) Exercise. $\square$

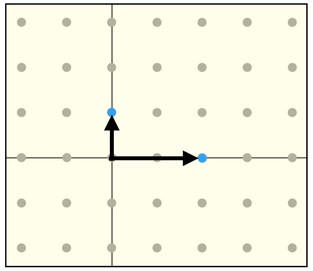
Volume of a lattice

- **Definition**. Let $L = L(B)$ be a lattice. The *volume* of L is $\mathrm{vol}(L) = |\det(B)|$.
- **Note:** The volume of a lattice is the "volume" of the fundamental parallelepiped of the lattice.
  - If the lattice is 2-dimensional, then its volume is the *area* of its parallelepiped.
  - Informally, the volume of a lattice is inversely proportional to the density of its lattice vectors. The larger the volume, the sparser is the lattice.
- **Exercise.** Show that the volume is an *invariant* of L, i.e., it doesn't depend on the basis B chosen for L.
- **Exercise**. Suppose that $L_1 \subseteq L_2$. Prove that $\mathrm{vol}(L_1) \geq \mathrm{vol}(L_2)$.

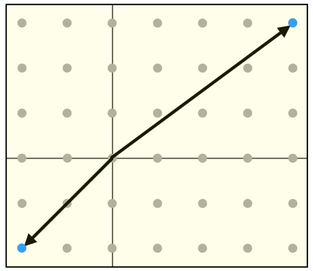
Some bases are nicer than others

- **Shortest Vector Problem (SVP)**: Given a lattice $L = L(B) \subseteq \mathbb{Z}^n$, find a shortest nonzero vector in L.
- **Example**: Consider the two SVP instances $L_2 = L(\{(2,0),(0,1)\})$ and $L_3 = L(\{(-2,-2),(4,3)\})$.
- So, hardness of an SVP instance $L(B)$ depends on the quality of the given basis B for L.

B_2



B_3



Successive minima

- A fundamental problem in lattice-based cryptanalysis is finding a "good" basis for a lattice.
- **Definition**: Let $L \subseteq \mathbb{Z}^n$ be a lattice. For each $i \in [1, n]$, the ith *successive minimum* $\lambda_i(L)$ is the smallest real number r such that L has i linearly independent vectors the longest of which has length r.
- **Notes**:

  1. $\lambda_1(L) \leq \lambda_2(L) \leq \cdots \leq \lambda_n(L)$.
  2. $\lambda_1(L)$ is the length of a shortest nonzero vector in L.
  3. (*Minkowski's Theorem*) $\lambda_1(L) \leq \sqrt{n}\,\mathrm{vol}(L)^{1/n}$.
  4. (*Gaussian Heuristic*) $\lambda_1(L) \approx \sqrt{n/(2\pi e)}\,\mathrm{vol}(L)^{1/n}$ for random lattices.
  5. $\lambda_n(L)$ is a lower bound on the length of a shortest basis for L.

LLL lattice basis reduction algorithm

- (1982) The **Lenstra-Lenstra-Lovász (LLL) algorithm** is a polynomial-time algorithm for finding a relatively short basis for a lattice L.
- **Notes**:

  1. The LLL algorithm is a clever modification of the Gram-Schmidt process for finding an orthogonal basis for a vector space in $\mathbb{R}^n$.
  2. Let $B = \{b_1, b_2, \ldots, b_n\}$ be the basis for L produced by the LLL algorithm, with $\|b_1\|_2 \leq \|b_2\|_2 \leq \cdots \leq \|b_n\|_2$. Then $\|b_i\|_2 \leq 2^{(n-1)/2}\lambda_i(L)$ for $1 \leq i \leq n$.
  In particular, $\|b_1\|_2 \leq 2^{(n-1)/2}\lambda_1(L)$ and $\|b_n\|_2 \leq 2^{(n-1)/2}\lambda_n(L)$.
  3. Also, $\|b_1\|_2 \leq 2^{(n-1)/4}\mathrm{vol}(L)^{1/n}$, and $\prod_{i=1}^{n} \|b_i\|_2 \leq 2^{n(n-1)/4}\mathrm{vol}(L)$.

Cryptanalytic applications of LLL

- Let $B = \{b_1, b_2, \ldots, b_n\}$ be the basis for L produced by the LLL algorithm, with $\|b_1\|_2 \leq \|b_2\|_2 \leq \cdots \leq \|b_n\|_2$. Then $\|b_i\|_2 \leq 2^{(n-1)/2}\lambda_i(L)$ for $1 \leq i \leq n$.
- In practice, the basis produced by LLL is typically significantly shorter than the above guarantee.
- LLL has been used to design attacks on many number-theoretic problems and public-key cryptographic systems.
  - e.g., see "Lattice attacks on digital signatures schemes", *Designs, Codes and Cryptography*, by N. Howgrave-Graham and N. Smart (2000): Finds the DSA (and ECDSA) secret key when a small number of bits of each per-message secret for several signed messages are leaked.
  - e.g., see "Lattice reduction in cryptology: an update", *Proceedings of ANTS-IV*, by P. Nguyen and J. Stern (2000).

SVP: A fundamental lattice problem

- **Shortest Vector Problem (SVP)**: Given a lattice $L = L(B)$, find a lattice vector of length $\lambda_1(L)$.
  - SVP is **NP-hard**.
  - The fastest (classical) algorithm known for SVP has (heuristic) running time $2^{0.292n+o(n)}$.
  - The fastest quantum algorithm known for SVP has (heuristic) running time $2^{0.265n+o(n)}$.
- **Approximate-SVP problem (SVP$_\gamma$)**: Given a lattice $L = L(B)$, find a nonzero lattice vector of length at most $\gamma \cdot \lambda_1(L)$.
  - SVP$_\gamma$ is believed to be hard for small γ.
  It's NP-hard for constant γ, but likely isn't NP-hard if $\gamma > \sqrt{n}$.
  - For $\gamma = 2^k$, the fastest algorithm known for SVP$_\gamma$ has running time $2^{\tilde{\Theta}(n/k)}$ (where $\tilde{\Theta}$ hides a power of $\log n$).
  - If $\gamma > 2^{(n \log\log n)/\log n}$, then SVP$_\gamma$ can be efficiently solved using the LLL algorithm.

SIVP: Another fundamental lattice problem

- **Shortest Independent Vectors Problem (SIVP)**: Given a lattice $L = L(B)$, find n linearly independent vectors in L all of which have length at most $\lambda_n(L)$.
  - A solution to SIVP isn't necessarily a basis for L.
  - SIVP is **NP-hard**.
- **Approximate-SIVP problem (SIVP$_\gamma$)**: Given a lattice $L = L(B)$, find n linearly independent vectors in L all of which have length at most $\gamma \cdot \lambda_n(L)$.
  - The hardness of SIVP$_\gamma$ is similar to that of SVP$_\gamma$.
  - **Fact**: SIVP$_{\gamma\sqrt{n}} \leq$ SVP$_\gamma$.

THE MATHEMATICS OF LATTICE-BASED CRYPTOGRAPHY

5. SIS/LWE and Lattices

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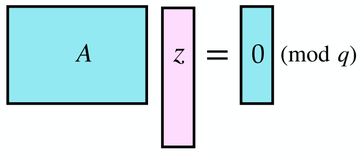
December 2024

Outline

1. The SIS lattice
2. Average-case hardness of SIS
3. The LWE lattice
4. Average-case hardness of LWE

Definition of the SIS lattice

- **SIS(n, m, q, B)**. Given $A \in_R \mathbb{Z}_q^{n\times m}$ (where $m \gg n$) and $B \ll q/2$, find $z \in \mathbb{Z}_q^m$ such that $Az = 0 \pmod q$, where $z \neq 0$ and $z \in [-B, B]^m$.
- Define the *SIS lattice* to be $L_A^\perp = \{z \in \mathbb{Z}^m : Az = 0 \pmod q\}$.



- **Claim 1.** $L_A^\perp$ is an integer lattice in $\mathbb{R}^m$.
- The claim can be easily proven using the following *equivalent* definition of a lattice.
- **Fact**. A lattice L is a discrete additive subgroup of $\mathbb{R}^m$.
  - L is an *additive subgroup* of $\mathbb{R}^m$ means that (i) L is non-empty subset of $\mathbb{R}^m$; and (ii) $x + y, -x \in L$ for all $x, y \in L$.
  - L is *discrete* means that for each $x \in L$, there exists $\epsilon > 0$ such that no element of L (other than x) is within distance ϵ of x.

Rank of the SIS lattice

- **Claim 2**. The SIS lattice $L_A^\perp = \{z \in \mathbb{Z}^m : Az = 0 \pmod q\}$ has full rank m.
- **Proof**. The lattice $q\mathbb{Z}^m$ is a sublattice of $L_A^\perp$.
  Now, the m vectors $(q,0,\ldots,0)$, $(0,q,\ldots,0)$, … , $(0,0,\ldots,q)$ are in $q\mathbb{Z}^m$ and are linearly independent (over $\mathbb{R}$).
  Thus, $q\mathbb{Z}^m$ is a full-rank lattice, and so $L_A^\perp$ is also a full-rank lattice. $\square$
- **Notes**.
  1. $L_A^\perp$ is a q-ary lattice, i.e. for all $z \in \mathbb{Z}^m$ we have $z \in L_A^\perp$ if and only if $z \bmod q \in L_A^\perp$.
  2. A basis matrix for the lattice $q\mathbb{Z}^m$ is qI_m.
     Thus, $\mathrm{vol}(q\mathbb{Z}^m) = |\det(qI_m)| = q^m$ and hence $\mathrm{vol}(L_A^\perp) \le q^m$.

Volume of the SIS lattice

- **Claim 3**. The SIS lattice $L_A^\perp = \{z \in \mathbb{Z}^m : Az = 0 \pmod q\}$ has volume q^n (assuming that A has rank n over $\mathbb{Z}_q$.)
- **Proof**. $\mathbb{Z}^m$ and $L_A^\perp$ are free (additive) abelian groups of rank m.
- Since $L_A^\perp$ is a subgroup of $\mathbb{Z}^m$, and they have the same rank, the quotient group $\mathbb{Z}^m/L_A^\perp$ is finite. Moreover, $\mathrm{vol}(L_A^\perp) = |\mathbb{Z}^m/L_A^\perp|$.
  (This is Theorem 1.17 in Stewart & Tall's book.)

See Section 1.6 of *Algebraic Number Theory and Fermat's Last Theorem* (3rd edition), by Stewart and Tall.

- So, to determine $\mathrm{vol}(L_A^\perp)$, we need to compute $|\mathbb{Z}^m/L_A^\perp|$, the number of cosets of $L_A^\perp$ in $\mathbb{Z}^m$.
  - Now, let $x, y \in \mathbb{Z}^m$. Then $L_A^\perp + x = L_A^\perp + y \iff x - y \in L_A^\perp \iff A(x-y) = 0 \pmod q \iff Ax = Ay \pmod q$.
  - Assuming that A has rank n over $\mathbb{Z}_q$, its column space has dimension n over $\mathbb{Z}_q$.
  - Thus, the column space of A has size q^n, whence $|\mathbb{Z}^m/L_A^\perp| = q^n$. $\square$

A basis of the SIS lattice

- **Claim 4**. Suppose that the first n columns of A are linearly independent over $\mathbb{Z}_q$, so A can be row-reduced to a matrix $\tilde{A} = [I_n | \overline{A}]$ (where $\overline{A} \in \mathbb{Z}_q^{n\times(m-n)}$).
  Then $C = \begin{bmatrix} qI_n & -\overline{A} \\ 0 & I_{m-n} \end{bmatrix} \in \mathbb{Z}^{m\times m}$ is a basis matrix for the SIS lattice $L_A^\perp$.
- **Proof**. Since A and $\tilde{A}$ are row equivalent (over $\mathbb{Z}_q$), they have the same null space (mod q). Hence, $L_{\tilde{A}}^\perp = L_A^\perp$, so we will find a basis for $L_{\tilde{A}}^\perp$.
  Now, each column v of C is in $L_{\tilde{A}}^\perp$ since $\tilde{A}v = 0 \pmod q$ [check this!].
  Moreover, the columns of C are linearly independent over $\mathbb{R}$ since $\det(C) = q^n$.
  Thus, C is a basis matrix for a full-rank sublattice L of $L_{\tilde{A}}^\perp$.
  Since $\mathrm{vol}(L) = q^n = \mathrm{vol}(L_A^\perp) = \mathrm{vol}(L_{\tilde{A}}^\perp)$, we have $L_{\tilde{A}}^\perp = L$.
  Thus, C is a basis matrix for the SIS lattice $L_A^\perp$. $\square$

Solving SIS

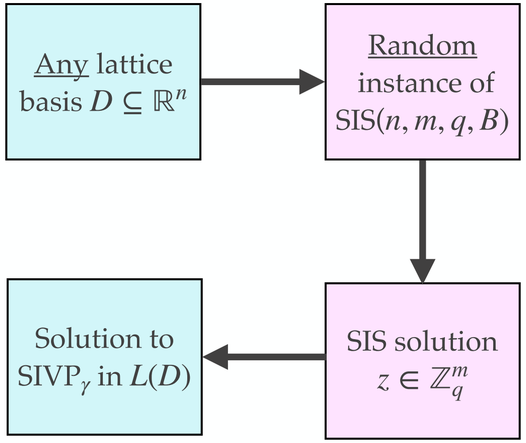
- **SIS(n, m, q, B)**. Given $A \in_R \mathbb{Z}_q^{n\times m}$ find $z \in \mathbb{Z}_q^m$ such that $Az = 0 \pmod q$, where $z \ne 0$ and $z \in [-B, B]^m$.
- An equivalent **lattice formulation** is:
  SIS(n, m, q, B): Given $A \in_R \mathbb{Z}_q^{n\times m}$, find a nonzero $z \in [-B, B]^m$ in the SIS lattice $L_A^\perp = L(C)$ where $C = \begin{bmatrix} qI_n & -\overline{A} \\ 0 & I_{m-n} \end{bmatrix}$.
- For $z \in \mathbb{R}^m$, the infinity norm of z is $\|z\|_\infty = \max_i |z_i|$.
  - So, an SIS solution $z \in \mathbb{Z}^m$ must satisfy $0 < \|z\|_\infty \le B$.
- SIS hardness is usually studied using the Euclidean norm: $\|z\|_2 = \sqrt{z_1^2 + z_2^2 + \cdots + z_m^2}$.
- **Exercise**: Show that for all $z \in \mathbb{R}^m$, $\|z\|_\infty \le \|z\|_2 \le \sqrt{m}\,\|z\|_\infty$.

Solving SIS_2

- **$\mathrm{SIS}_2(n, m, q, \beta)$**. Given $A \in_R \mathbb{Z}_q^{n \times m}$ where $\beta \ll q$, find nonzero $z \in \mathbb{Z}_q^m$ such that $Az = 0 \pmod q$ and $\|z\|_2 \le \beta$.
- An equivalent lattice formulation is:
  $\mathrm{SIS}_2(n, m, q, \beta)$: Given $A \in_R \mathbb{Z}_q^{n \times m}$, find nonzero z with $\|z\|_2 \le \beta$ in the SIS lattice $L_A^\perp$.
  - By Minkowski's Theorem (slide 49), $\lambda_1(L_A^\perp) \le \sqrt{m}\, q^{n/m}$.
  - We'll assume that $\beta \ge \sqrt{m}\, q^{n/m}$, whereby an SIS_2 solution is guaranteed to exist.
- Now, by the Gaussian heuristic (slide 49), $\lambda_1(L_A^\perp) \approx \sqrt{m/(2\pi e)}\, q^{n/m}$.
- Thus, SIS_2 can be seen as an instance of approximate-SVP (SVP_γ) in the SIS lattice $L_A^\perp$ with approximation factor $\gamma = \beta\sqrt{2\pi e}/(\sqrt{m} q^{n/m})$.
- **Exercise**: Show that $\mathrm{SIS}(n, m, q, B) \le \mathrm{SIS}_2(n, m, q, B) \le \mathrm{SIS}(n, m, q, B/\sqrt{m})$.

Average-case hardness of SIS

- It's reasonable to conjecture that SIS is hard in the *worst case*.
- But, what can we say about the hardness of SIS *on average*?
- In 1996, Ajtai proved a striking *average case hardness result* for SIS:
  - If SIVP_γ is hard in the *worst-case*, then SIS is hard on *average*.
  - Such a reduction is called a *worst-case to average-case reduction*.
- Since the assumption that SIVP_γ is hard in the worst case is a reasonable assumption, we have a provable guarantee that SIS is hard on average.



The worst-case to average-case reduction is asymptotic

- Although Ajtai's worst-case to average-case reduction provides a strong guarantee for the average-case hardness of SIS, the guarantee is an *asymptotic* one.
  - Also, the reduction is *highly non-tight*.
- In 2004, Micciancio & Regev proved the following:
  Theorem. For any $m(n) = \Theta(n \log n)$, there exists a $q(n) = O(n^2 \log n)$ such that for any function $\gamma(n) = \omega(n \log n)$, solving $\mathrm{SIS}_2(n, m, q, \beta)$ on average with non-negligible probability is at least as hard as solving SIVP_γ in the worst case.

WORST-CASE TO AVERAGE-CASE REDUCTIONS BASED ON GAUSSIAN MEASURES*

DANIELE MICCIANCIO† AND ODED REGEV‡

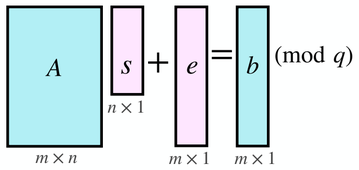
SIS summary

SIS is considered a lattice problem for two reasons.

1. SIS is equivalent to solving SVP_γ in the SIS lattice.
   - The fastest algorithm known for solving SVP_γ is the Block-Korkine-Zolotarev (BKZ) algorithm, which has an exponential running time.
   - The running time of BKZ is used to select concrete parameters for SIS for a desired security level.
2. Solving SIS on average is provably at least as hard as solving SIVP_γ in the worst case.
   - This hardness guarantee is an asymptotic one, and its relevance to the hardness SIS in practice is not clear.

Definition of the LWE lattice

- **LWE(m, n, q, B)**. Let $s \in_R \mathbb{Z}_q^n$ and $e \in_R [-B, B]^m$. Given $A \in_R \mathbb{Z}_q^{m \times n}$ and $b = As + e \pmod q$, find s.



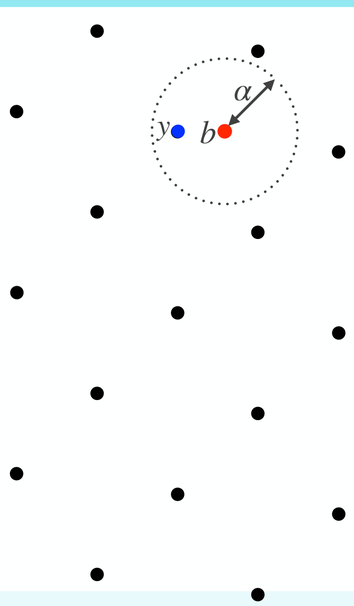
- Define the *LWE lattice* to be $L_A = \{y \in \mathbb{Z}^m : Az = y \pmod q \text{ for some } z \in \mathbb{Z}^n\} \subseteq \mathbb{R}^m$.
- **Claim 1.** L_A is a full-rank (integer) q-ary lattice in $\mathbb{R}^m$.
- **Proof**. L_A is a discrete additive subgroup of $\mathbb{R}^m$, and thus is a lattice. Since $q\mathbb{Z}^m$ is a sublattice of L_A, it follows that L_A is q-ary and has rank m. $\square$

A basis of the LWE lattice

- **Claim 2.** Let $A = \begin{bmatrix} A_1 \\ A_2 \end{bmatrix}$ where $A_1 \in \mathbb{Z}_q^{n \times n}$ and $A_2 \in \mathbb{Z}_q^{(m-n) \times n}$, and suppose that A_1 is invertible mod q. Let $D_2 = A_2 A_1^{-1} \pmod q$. Then $D = \begin{bmatrix} I_n & 0 \\ D_2 & qI_{m-n} \end{bmatrix} \in \mathbb{Z}^{m \times m}$ is a basis matrix for L_A (and so $\mathrm{vol}(L_A) = q^{m-n}$).
- **Proof**. Since $\det(D) = q^{m-n}$, the columns of D are linearly independent over $\mathbb{R}$. Write $y \in \mathbb{Z}^m$ as $\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ where $y_1 \in \mathbb{Z}^n$ and $y_2 \in \mathbb{Z}^{m-n}$.
  Now, $y \in L_A \iff y = Az \pmod q$ for some $z \in \mathbb{Z}^n \iff y_1 = A_1 z \pmod q$ and $y_2 = A_2 z \pmod q$ for some $z \in \mathbb{Z}^n \iff y_2 = A_2 A_1^{-1} y_1 \pmod q \iff y_2 = D_2 y_1 + qc$ for some $c \in \mathbb{Z}^{m-n}$.
  Observing that $y = D \begin{bmatrix} y_1 \\ c \end{bmatrix}$, it follows that the columns of D are a basis for L_A. $\square$

Solving LWE

- **LWE(m, n, q, B)**. Let $s \in_R \mathbb{Z}_q^n$ and $e \in_R [-B, B]^m$. Given $A \in_R \mathbb{Z}_q^{m \times n}$ and $b = As + e \pmod q$, find s.
- **LWE lattice**: $L_A = \{y \in \mathbb{Z}^m : As = y \pmod q \text{ for some } s \in \mathbb{Z}^n\} \subseteq \mathbb{R}^m$.
- Note that for an LWE instance (A, b, s, e), we have $y = As \bmod q \in L_A$, and $\|b - y\|_2 = \|e\|_2 \le \sqrt{m}\, B$.
- Thus, LWE is a special instance of the following lattice problem:
  Bounded Distance Decoding (BDD$_\alpha$):
  Given a lattice $L = L(D) \subseteq \mathbb{R}^m$ and $b \in \mathbb{R}^m$ with the guarantee that there is a unique $y \in L$ within distance α of b, find y.



Reducing BDD to SVP (1)

- **BDD$_\alpha$**: Given a lattice $L = L(D) \subseteq \mathbb{R}^m$ and $b \in \mathbb{R}^m$ with the guarantee that there is a unique $y \in L$ within distance α of b, find y.
- We'll suppose that $\alpha < \lambda_1(L)/\sqrt{2}$.
- Let $D' = \begin{bmatrix} D & -b \\ 0 & \alpha \end{bmatrix} \in \mathbb{R}^{(m+1) \times (m+1)}$. Then $L' = L(D') = \left\{ \begin{bmatrix} v - cb \\ c\alpha \end{bmatrix} : v \in L(D) \text{ and } c \in \mathbb{Z} \right\}$.
- Notice that for $(v, c) = (y, 1)$, we have $\tilde{v} = \begin{bmatrix} y - b \\ \alpha \end{bmatrix} \in L'$ with $\|\tilde{v}\|_2 = \sqrt{\|y - b\|_2^2 + \alpha^2} \le \sqrt{2}\alpha$. Hence, $\lambda_1(L') \le \sqrt{2}\alpha < \lambda_1(L)$.
- Suppose now that $v' = \begin{bmatrix} v - cb \\ c\alpha \end{bmatrix} \in L'$ has length $\|v'\|_2 = \lambda_1(L')$.
- If $c = 0$, then $\|v'\|_2 = \|v\|_2 \ge \lambda_1(L) > \lambda_1(L')$, a contradiction.
- And, if $|c| \ge 2$, then $\|v'\|_2 \ge 2\alpha > \sqrt{2}\alpha \ge \lambda_1(L')$, a contradiction.
- Hence, we must have $c = \pm 1$. If $c = 1$, we have $v' = \begin{bmatrix} v - b \\ \alpha \end{bmatrix}$ for some $v \in L$.
- Now, if $v \ne y$, then $\|v - b\|_2 > \|y - b\|_2$, whence $\|v'\|_2 > \|\tilde{v}\|_2$, contradicting $\|v'\|_2 = \lambda_1(L')$.
- Hence $\pm\tilde{v}$ are the only vectors of length $\lambda_1(L')$ in L'.

Reducing BDD to SVP (2)

- **BDD**$_\alpha$: Given a lattice $L = L(D) \subseteq \mathbb{R}^m$ and $b \in \mathbb{R}^m$ with the guarantee that there is a unique $y \in L$ within distance α of b, find y.
- **Summary**: We can solve the BBD$_\alpha$ instance by solving SVP for $L(D')$ where $D' = \begin{bmatrix} D & -b \\ 0 & \alpha \end{bmatrix}$.
- This method of solving LWE is called a "primal attack using a Kannan embedding".

Average-case hardness of LWE

- It's reasonable to conjecture that LWE is hard in the *worst case*.
- But, what can we say about the hardness of LWE *on average*?
- In 2005, Regev proved a striking *average-case hardness result* for LWE:
  - If SIVP$_\gamma$ is quantumly hard in the *worst-case*, then LWE is hard on *average*.
- Since the assumption that SIVP$_\gamma$ is quantumly hard in the worst case is a reasonable assumption, we have a provable guarantee that LWE is hard on average.
- However, as with Ajtai's worst-case to average-case reduction for SIS, Regev's reduction is *highly non-tight* (and also a quantum reduction).
  - For a concrete analysis of Regev's reduction, see Section 5 of: "Another look at tightness II: practical issues in cryptography" by Chatterjee, Koblitz, Menezes & Sarkar, https://eprint.iacr.org/2016/360.

Gaussian distributions

- I should note that in Regev's worst-case to average-case reduction, and also in much of the cryptographic literature on LWE-based protocols, the components of the LWE error vector e are drawn from certain Gaussian distributions (and not from uniform distributions)
- However, for the sake of simplicity, I didn't use Gaussians in my lectures.
- Also, Kyber and Dilithium use uniform distributions and central binomial distributions.

LWE summary

LWE is considered a lattice problem for two reasons.

1. LWE can be reduced to solving BDD$_\alpha$ in the LWE lattice, which in turn can be reduced to solving an instance of SVP.
   - The fastest algorithm known for solving SVP is the Block-Korkine-Zolotarev (BKZ) algorithm, which has an exponential running time.
   - The running time of BKZ can be used to select concrete parameters for LWE for a desired security level.

2. Solving LWE on average is provably at least as hard as (quantumly) solving SIVP$_\gamma$ in the worst case.
   - This hardness guarantee is an asymptotic one, and its relevance to LWE in practice is not clear.

THE MATHEMATICS OF LATTICE-BASED CRYPTOGRAPHY

6. Ring-SIS and Ring-LWE

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Outline

1. Polynomials rings
2. Ideal lattices
3. Cyclic lattices
4. Anti-cyclic lattices
5. Ring-SIS
6. Ring-LWE

LWE-based public-key encryption

Key generation: Alice does:

1. Select $s \in_R [-B, B]^n$.
2. Select $A \in_R \mathbb{Z}_q^{n \times n}$ and $e \in_R [-B, B]^n$.
3. Compute $b = As + e$.
4. Alice's **public key** is (A, b); her **private key** is s.

Encryption: To encrypt a message $m \in \{0,1\}$ for Alice, Bob does:

1. Obtain an authentic copy of Alice's encryption key (A, b).
2. Select $r, z \in_R [-B, B]^n$ and $z' \in_R [-B, B]$.
3. Compute $c_1 = A^T r + z$ and $c_2 = b^T r + z' + m\lceil q/2 \rfloor$.
4. Output $c = (c_1, c_2)$.

Decryption: To decrypt $c = (c_1, c_2)$, Alice does:

1. Output 0 if $|c_2 - s^T c_1| < q/4$, and 1 otherwise.

Module-LWE: Replace $\mathbb{Z}_q$ elements with polynomials in a certain polynomial ring over $\mathbb{Z}_q$.

Polynomial rings

- $\mathbb{Z}[x]$ is the set of polynomials in x with integer coefficients.
- Let $f \in \mathbb{Z}[x]$ be a *monic* polynomial of degree n.
- The **polynomial ring** $R = \mathbb{Z}[x]/(f)$ is comprised of the set of all polynomials in $\mathbb{Z}[x]$ of degree less than n, with multiplication of polynomials performed modulo the **reduction polynomial** $f(x)$.
- So, to multiply polynomials $a(x), b(x) \in R$:
  1. Multiply $a(x)$ and $b(x)$ in $\mathbb{Z}[x]$, obtaining a polynomial $h(x)$ of degree at most $2n - 2$.
  2. Divide $h(x)$ by $f(x)$ to get a remainder polynomial $r(x)$ of degree at most $n - 1$.
  3. Then $a(x) \times b(x) = r(x)$ in R.

Representing polynomials as vectors

- A polynomial $a(x) = a_0 + a_1x + \cdots + a_{n-1}x^{n-1}$ in $R = \mathbb{Z}[x]/(f)$ can be represented by its vector of coefficients $a = (a_0, a_1, \ldots, a_{n-1})$. The vector has length exactly n.
- I'll write $a(x) \leftrightarrow a$.
- **Example**: Consider $R = \mathbb{Z}[x]/(x^4 + 2x^2 - 11x + 1)$.
  - The polynomials $a(x) = 23 + 11x^2 + 7x^3 \in R$ and $b(x) = 40 + 5x + 16x^2 \in R$ can be represented by the vectors $a = (23, 0, 11, 7)$ and $b = (40, 5, 16, 0)$
  - In R, we have $a + b = (63, 5, 27, 7)$, $a - b = (-17, -5, -5, 7)$, and $a \times b = (709, 2324, 1618, 111)$.

Ideals

- Let $R = \mathbb{Z}[x]/(f)$.
- An **ideal** of R is a subset $I \subseteq R$ such that:
  i) $0 \in I$.
  ii) If $a, b \in I$ then $a + b \in I$ and $a - b \in I$.
  iii) If $a \in I$ and $r \in R$, then $a \times r \in I$.
- **Example**: Let $a(x) \in R$. Then $\langle a(x) \rangle = \{a(x)r(x) \bmod f(x) \mid r \in R\}$ is an ideal of R, called the **principal ideal** generated by $a(x)$.
- **Example**: Let $a_1(x), a_2(x) \in R$. Then $\langle a_1(x), a_2(x) \rangle = \{a_1(x)r_1(x) + a_2(x)r_2(x) \bmod f(x) \mid r_1, r_2 \in R\}$ is an ideal of R, called the ideal generated by $a_1(x)$ and $a_2(x)$.

Ideal lattices

- Let $R = \mathbb{Z}[x]/(f)$ and let I be a nonzero ideal of R.
- Then $L(I) = \{a \mid a(x) \in I\}$ is an integer lattice in $\mathbb{R}^n$, called an **ideal lattice**.
- **Remark**. $L(I)$ does not necessarily have rank n, i.e., $L(I)$ might be spanned by $n - 1$ or fewer linearly independent lattice vectors.
- We'll only be concerned with the cases $f(x) = x^n - 1$ (*cyclic lattices*) and $f(x) = x^n + 1$ (*anti-cyclic lattices*).

Cyclic lattices

- A lattice L is said to be **cyclic** if $v \in L$ implies that the right cyclic shift of v is also in L. (The *right cyclic shift* of $v = (v_0, v_1, \ldots, v_{n-1})$ is $(v_{n-1}, v_0, v_1, \ldots, v_{n-2})$.)
- Cyclic lattices are examples of **structured lattices**.
- Cyclic lattices were first studied by Micciancio in 2002.
- **Claim**. Let $R = \mathbb{Z}[x]/(x^n - 1)$. Then every ideal lattice is cyclic.
- **Proof**. Let $L = L(I)$ be an ideal lattice, and let $v \in L$.
  Recall that $v = (v_0, v_1, \ldots, v_{n-1}) \leftrightarrow v(x) = v_0 + v_1x + \cdots + v_{n-1}x^{n-1}$.
  Now, $xv(x) = v_0x + v_1x^2 + \cdots + v_{n-1}x^n$
  $$= v_{n-1} + v_0x + v_1x^2 + \cdots + v_{n-2}x^{n-1} \bmod (x^n - 1)$$
  $$\leftrightarrow (v_{n-1}, v_0, v_1, \ldots, v_{n-2}),$$
  which is in L since I is an ideal (whence $xv(x) \in I$). $\square$

Matrix representation of a cyclic lattice (1)

- Let $R = \mathbb{Z}[x]/(x^n - 1)$, and let $a(x) \in R$.
- Let $I = \langle a(x) \rangle$, and consider $L = L(I)$.
- Now, $I = \{a(x)r(x) \bmod (x^n - 1) \mid r(x) \in R\}$,
- If $r(x) = r_0 + r_1 x + \cdots + r_{n-1}x^{n-1}$, then
  $a(x)r(x) = r_0 a(x) + r_1 x a(x) + \cdots + r_{n-1}x^{n-1}a(x) \bmod (x^n - 1)$.
- Hence, $\{a(x), xa(x), x^2a(x), \ldots, x^{n-1}a(x)\}$ is a spanning set for L (all polynomials are modulo $x^n - 1$).
- More precisely, the set of vector representations of
  $a(x),\ xa(x) \bmod (x^n - 1),\ \ldots,\ x^{n-1}a(x) \bmod (x^n - 1)$
  is a spanning set for L.

Matrix representation of a cyclic lattice (2)

- Let A be the $n \times n$ matrix whose columns are the vector representations of $a(x), xa(x), \ldots, x^{n-1}a(x) \bmod (x^n - 1)$.
- So, $A = \begin{bmatrix} a_0 & a_{n-1} & a_{n-2} & \cdots & a_1 \\ a_1 & a_0 & a_{n-1} & \cdots & a_2 \\ a_2 & a_1 & a_0 & \cdots & a_3 \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \cdots & a_0 \end{bmatrix}$.
- A is a **circulant matrix**, denoted $A = \mathrm{circ}(a)$.
- **Fact**. A is invertible, i.e., $L(A)$ is a full-rank lattice, if and only if $\gcd(a(x), x^n - 1) = 1$ over $\mathbb{Q}$.

- Let's henceforth assume that $\gcd(a(x), x^n - 1) = 1$.
- Now, if $r = (r_0, r_1, \ldots, r_{n-1})^T \in \mathbb{Z}^n$, then

$$Ar = r_0 \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{n-1} \end{bmatrix} + r_1 \begin{bmatrix} a_{n-1} \\ a_0 \\ a_1 \\ \vdots \\ a_{n-2} \end{bmatrix} + r_2 \begin{bmatrix} a_{n-2} \\ a_{n-1} \\ a_0 \\ \vdots \\ a_{n-3} \end{bmatrix} + \cdots + r_{n-1} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_0 \end{bmatrix}$$

$$\begin{aligned} &\leftrightarrow r_0 a(x) + r_1 x a(x) + r_2 x^2 a(x) + \cdots + r_{n-1}x^{n-1}a(x) \bmod (x^n - 1) \\ &= a(x)(r_0 + r_1 x + r_2 x^2 + \cdots + r_{n-1}x^{n-1}) \bmod (x^n - 1) \\ &= a(x)r(x) \bmod (x^n - 1)\,. \end{aligned}$$

- Thus, $L(\langle a(x) \rangle) = \{a(x)r(x) \bmod (x^n - 1) \mid r(x) \in R\}$
  $\leftrightarrow \{Ar \mid r \in \mathbb{Z}^n\}$
  $= L(A)$.

Matrix representation of a cyclic lattice (3)

- **Summary**. Let $R = \mathbb{Z}[x]/(x^n - 1)$, and let $a(x) \in R$ with $\gcd(a(x), x^n - 1) = 1$ over $\mathbb{Q}$. Then $L(\langle a(x) \rangle) = L(A)$ where $A = \mathrm{circ}(a)$.
- More generally, let $a_1, a_2, \ldots, a_\ell \in R$ with $\gcd(a_i, x^n - 1) = 1$ over $\mathbb{Q}$.
- Let $I = \langle a_1(x), a_2(x), \ldots, a_\ell(x) \rangle$ be an ideal in R.
- Recall that $I = \{a_1 r_1 + a_2 r_2 + \cdots + a_\ell r_\ell \mid r_i \in R\}$.
- Let $m = \ell n$, and define the $n \times m$ matrix
  $A = [A_1 | A_2 | \cdots | A_\ell]$, where $A_i = \mathrm{circ}(a_i)$.
- Then, for $r_1, r_2, \ldots, r_\ell \in \mathbb{Z}^n$,
  $A_1 r_1 + A_2 r_2 + \cdots + A_\ell r_\ell$
  $\leftrightarrow a_1(x)r_1(x) + a_2(x)r_2(x) + \cdots + a_\ell(x)r_\ell(x) \pmod{(x^n - 1)}$.
- Hence $L(I) = L(A)$ is a rank-n, cyclic, integer lattice in $\mathbb{R}^n$.

$A_i = \mathrm{circ}(a_i)$

$$A = \boxed{A_1 \mid A_2 \mid \cdots \mid A_\ell}$$

One-way function

- **Setup**. Let $R = \mathbb{Z}[x]/(x^n - 1)$, let $a_1, a_2, \ldots, a_\ell \in_R R$, and let $m = \ell n$.
  Let $A = [A_1 | A_2 | \cdots | A_\ell]$ where $A_i = \mathrm{circ}(a_i)$.
  Let q be a (relatively small) prime number with $2^m > q^n$.
- Consider the compression function $H_A : \{0,1\}^m \longrightarrow \mathbb{Z}_q^n$ defined by $H_A(z) = Az \pmod{q}$.
- Micciancio (2002) proved that H_A is a **one-way function** *on average*, provided that a certain lattice problem in cyclic lattices is hard in the *worst-case*.
  (The problem is to approximate the "covering radius" of any cyclic lattice.)
- However, H_A is *not* collision resistant.

$$A = \boxed{A_1 \mid A_2 \mid \cdots \mid A_\ell}$$

Finding collisions for H_A

- **Recall**. $H_A : \{0,1\}^m \longrightarrow \mathbb{Z}_q^n$ is defined by $H_A(z) = Az \pmod q$, where $A = [A_1 | A_2 | \cdots | A_\ell]$ and $A_i = \text{circ}(a_i)$.

$$A = \boxed{A_1 \mid A_2 \mid \cdots \mid A_\ell}$$

- The problem is that since each A_i is circulant, we have $A_i \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} c_i \\ \vdots \\ c_i \end{bmatrix}$,

  where c_i is the sum of the entries in the first row (or first column) of A_i.
- This observation can be used to efficiently find a collision for H_A as follows:
  1. Determine $x_1, x_2, \ldots, x_\ell \in \{0, \pm 1\}$, such that $x_1c_1 + x_2c_2 + \cdots + x_\ell c_\ell = 0 \pmod q$. At least one of the x_i's should be 1, and at least one should be -1.
  2. Define z to be the length-m vector $[z_1, z_2, \ldots, z_\ell]^T$, where z_i is the length-n vector all of whose components are x_i. Note that $Az = 0 \pmod q$.
  3. Let r_1 be the length-m vector obtained from z by setting all -1 entries to 0, and let $r_2 = r_1 - z$. Then (r_1, r_2) is a collision for H_A.

Anti-cyclic lattices

- A lattice L is said to be **anti-cyclic** if $v = (v_0, v_1, \ldots, v_{n-1}) \in L$ implies that $(-v_{n-1}, v_0, v_1, \ldots, v_{n-2})$ is also in L.
- Anti-cyclic lattices were first studied by Lyubashevsky & Micciancio in 2006.
- Anti-cyclic lattices are **structured lattices.**
- **Claim**. Let $R = \mathbb{Z}[x]/(x^n + 1)$. Then every ideal lattice is anti-cyclic.
- **Proof**. Let $L = L(I)$ be an ideal lattice, and let $v \in L$.
  Recall that $v = (v_0, v_1, \ldots, v_{n-1}) \leftrightarrow v(x) = v_0 + v_1x + \cdots + v_{n-1}x^{n-1}$.
  Now, $xv(x) = v_0x + v_1x^2 + \cdots + v_{n-1}x^n$

$$= -v_{n-1} + v_0x + v_1x^2 + \cdots + v_{n-2}x^{n-1} \bmod (x^n + 1)$$
$$\leftrightarrow (-v_{n-1}, v_0, v_1, \ldots, v_{n-2}),$$

  which is in L since I is an ideal (whence $xv(x) \in I$). $\square$

Matrix representation of an anti-cyclic lattice (1)

- Let $n = 2^w$. Then $x^n + 1$ is irreducible over $\mathbb{Q}$.
- Let $R = \mathbb{Z}[x]/(x^n + 1)$, and let $a(x) \in R$ with $a(x) \neq 0$. Then $\gcd(a(x), x^n + 1) = 1$.
- Let $I = \langle a(x) \rangle$, and consider $L = L(I)$.
- Now, $I = \{a(x)r(x) \bmod (x^n + 1) \mid r(x) \in R\}$.
- Hence, the set of vector representations of $a(x), xa(x) \bmod (x^n + 1), \ldots$ $x^{n-1}a(x) \bmod (x^n + 1)$ is a spanning set for L.
- In fact, the vectors are a basis for L.

- Let A be the $n \times n$ matrix whose columns are the vector representations of $a(x), xa(x), \ldots, x^{n-1}a(x) \bmod (x^n + 1)$.
- So, $A = \begin{bmatrix} a_0 & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \\ a_1 & a_0 & -a_{n-1} & \cdots & -a_2 \\ a_2 & a_1 & a_0 & \cdots & -a_3 \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n-1} & a_{n-2} & a_{n-3} & \cdots & a_0 \end{bmatrix}$.
- Then $L(I) = L(A)$.
- A is an **anti-circulant matrix**, denoted $A = \overline{\text{circ}}(a)$.
- Notice that the row sums (and column sums) are no longer equal.

Example: Multiplication in $R = \mathbb{Z}[x]/(x^n + 1)$

- Let $n = 4$, so $R = \mathbb{Z}[x]/(x^4 + 1)$.
- Let $a(x) = 3 - 2x + 5x^2 - 4x^3 \in R$.
- Then $A = \overline{\text{circ}}(a) = \begin{bmatrix} 3 & 4 & -5 & 2 \\ -2 & 3 & 4 & -5 \\ 5 & -2 & 3 & 4 \\ -4 & 5 & -2 & 3 \end{bmatrix}$.
- Let $r(x) = 1 + 2x + 11x^2 - 7x^3 \in R$.
- **Exercise**: Check that $a(x) \times r(x) \pmod{x^4 + 1} = -58 + 83x + 6x^2 - 37x^3$, and $A \cdot r^T = A \cdot [1, 2, 11, -7]^T = [-58, 83, 6, -37]^T$.

Matrix representation of an anti-cyclic lattice (2)

- **Summary**. Let $n = 2^w$, let $R = \mathbb{Z}[x]/(x^n + 1)$, and let $a(x) \in R$ with $a(x) \neq 0$. Then $L(\langle a(x) \rangle) = L(A)$ where $A = \overline{\text{circ}}(a)$.

- More generally, let $a_1, a_2, \ldots, a_\ell \in R$ with $a_i \neq 0$.
- Let $I = \langle a_1(x), a_2(x), \ \ldots, a_\ell(x) \rangle$ be an ideal in R.
- Let $m = \ell n$, and define the $n \times m$ matrix
$$A = [A_1 \,|\, A_2 \,|\, \cdots \,|\, A_\ell],$$
where $A_i = \overline{\text{circ}}(a_i)$.

$A_i = \overline{\text{circ}}(a_i)$

$$A = \begin{array}{|c|c|c|c|} \hline A_1 & A_2 & \cdots & A_\ell \\ \hline \end{array}$$

- Then, $L(I) = L(A)$ is a rank-n, anti-cyclic, integer lattice in $\mathbb{R}^n$.
- These *structured lattices* can be used to define a special case of SIS, called **Ring-SIS**.

Sizes

- Recall: q prime, $\mathbb{Z}_q = \{0, 1, \ldots, q-1\}$, $n = 2^w$.
  - $R = \mathbb{Z}[x]/(x^n + 1)$, $R_q = \mathbb{Z}_q[x]/(x^n + 1)$.
- For $r \in \mathbb{Z}_q$, define $r \bmod q = \begin{cases} r, & \text{if } 0 \leq r \leq (q-1)/2, \\ q - r & \text{if } (q-1)/2 < r \leq q-1. \end{cases}$
- For $r \in \mathbb{Z}_q$, define $\|r\|_\infty = |\, r \bmod q \,|$.
  - **Example**: For $q = 23$, $\|3\|_\infty = 3$ and $\|19\|_\infty = 4$.
- For $a(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1} \in R_q$, define $\|a\|_\infty = \max \|a_i\|_\infty$.
  - **Example**: Let $q = 23$ and $n = 8$. Then $\|3 + 19x^2 + 21x^3 + x^4\|_\infty = 4$.

Ring-SIS

- Introduced by Lyubashevsky-Micciancio and Peikert-Rosen (2006).
- **Ring-SIS**(n, ℓ, q, B):
Given $a_1, a_2, \ldots, a_\ell \in_R R_q$, find $z_1, z_2, \ldots, z_\ell \in R_q$ such that $a_1 z_1 + a_2 z_2 + \cdots + a_\ell z_\ell = 0 \pmod{q}$, where $\|z_i\|_\infty \leq B$ and not all z_i are 0.
  - **Note**: If $(z_1, z_2, \ldots, z_\ell)$ is a solution then so is $(xz_1, xz_2, \ldots, xz_\ell)$.
- *Equivalently*, given $a_1, a_2, \ldots, a_\ell \in_R R_q$, find nonzero $z \in [-B, B]^m$ such that $Az = 0 \pmod{q}$, where $A = [\overline{\text{circ}}(a_1) \,|\, \cdots \,|\, \overline{\text{circ}}(a_\ell)]_{n \times m}$.
- So, Ring-SIS is a special case of SIS where the matrix A is *structured*.

$A_i = \overline{\text{circ}}(a_i)$

$$A = \begin{array}{|c|c|c|c|} \hline A_1 & A_2 & \cdots & A_\ell \\ \hline \end{array}$$

- Lyubashevsky and Micciancio proved that solving Ring-SIS on *average* is at least as hard as solving SVP_γ for anti-cyclic lattices *in the worst case*.

Example: Ring-SIS (1)

- Let $q = 59$, $n = 4$, $f(x) = x^4 + 1$, $R_q = \mathbb{Z}_{59}[x]/(x^4 + 1)$, $\ell = 3$, $B = 2$.
- Let $a_1(x) = 10 + 16x^2 + 51x^3$, $a_2(x) = 41 + 10x + 54x^2 + 16x^3$, $a_3(x) = 11 + 17x + 39x^2 + 5x^3 \in R_q$.
- **Ring-SIS instance:**
Find $z_1, z_2, z_3 \in R_q$, not all 0, with $a_1 z_1 + a_2 z_2 + a_3 z_3 = 0 \pmod{q}$ and $\|z_i\|_\infty \leq 2$.
- We have
$$A = \underbrace{\left[\begin{array}{cccc|cccc|cccc} 10 & 8 & 43 & 0 & 41 & 43 & 5 & 49 & 11 & 54 & 20 & 42 \\ 0 & 10 & 8 & 43 & 10 & 41 & 43 & 5 & 17 & 11 & 54 & 20 \\ 16 & 0 & 10 & 8 & 54 & 10 & 41 & 43 & 39 & 17 & 11 & 54 \\ 51 & 16 & 0 & 10 & 16 & 54 & 10 & 41 & 5 & 39 & 17 & 11 \end{array}\right]_{4 \times 12}}_{\overline{\text{circ}}(a_1) \quad \overline{\text{circ}}(a_2) \quad \overline{\text{circ}}(a_3)}.$$

Example: Ring-SIS (2)

- Gaussian elimination (mod q) on A yields the following matrix in reduced form:

$$A' = \begin{bmatrix} 1 & 0 & 0 & 0 & 48 & 6 & 43 & 45 & 7 & 3 & 58 & 57 \\ 0 & 1 & 0 & 0 & 14 & 48 & 6 & 43 & 2 & 7 & 3 & 58 \\ 0 & 0 & 1 & 0 & 16 & 14 & 48 & 6 & 1 & 2 & 7 & 3 \\ 0 & 0 & 0 & 1 & 53 & 16 & 14 & 48 & 56 & 1 & 2 & 7 \end{bmatrix}.$$

- The set of all solutions $r = (r_1, r_2, \ldots, r_{12}) \in \mathbb{Z}_{59}^{12}$ to $A'r = 0 \pmod q$ is:

$$\begin{aligned} r_1 &= 11r_5 + 53r_6 + 16r_7 + 14r_8 + 52r_9 + 56r_{10} + r_{11} + 2r_{12} \\ r_2 &= 45r_5 + 11r_6 + 53r_7 + 16r_8 + 57r_9 + 52r_{10} + 56r_{11} + r_{12} \\ r_3 &= 43r_5 + 45r_6 + 11r_7 + 53r_8 + 58r_9 + 57r_{10} + 52r_{11} + 56r_{12} \\ r_4 &= 6r_5 + 43r_6 + 45r_7 + 11r_8 + 3r_9 + 58r_{10} + 57r_{11} + 52r_{12}\,. \end{aligned}$$

Example: Ring-SIS (3)

- The total number of solutions to $A'r = 0 \pmod q$ is $q^8 = 146{,}830{,}437{,}604{,}321$.
  - Of these, the number of solutions r that are nonzero and in $[-2, 2]^{12}$ is 24.
- The nonzero Ring-SIS solutions (up to multiplication by $\pm 1,\ \pm x,\ \pm x^2,\ \pm x^3$) are:

$$\begin{aligned} R_1 &= (1, 2, -1, 2, -1, 2, 0, -2, 1, 0, 0, 0) \\ R_2 &= (1, 1, 0, 0, -2, -2, 0, 0, -2, 2, -1, 2) \\ R_3 &= (1, 2, -2, 1, -1, 2, 2, 0, 0, 2, 2, -2)\,. \end{aligned}$$

- For example, the first solution R_1 in polynomial form is:

$$z_1(x) = 1 + 2x - x^2 + 2x^3, \quad z_2(x) = -1 + 2x - 2x^3, \quad z_3(x) = 1.$$

- **Check**: $AR_1 = 0 \pmod q$ and $a_1(x)z_1(x) + a_2(x)z_2(x) + a_3(x)z_3(x) = 0$ in R_q.

Collision-resistant hash function

- **Setup**. Select q and ℓ with $\ell > \log q$.
  Let $R_q = \mathbb{Z}_q[x]/(x^n + 1)$, let $a_1, a_2, \ldots, a_\ell \in_R R_q$, and let $m = \ell n$.
  Let $A = [A_1 \,|\, A_2 \,|\, \cdots \,|\, A_\ell]$ where $A_i = \overline{\mathrm{circ}}(a_i)$.

$A_i = \overline{\mathrm{circ}}(a_i)$

$A =$	A_1	A_2	$\cdots$	A_ℓ

- Consider the compression function $H_A : \{0,1\}^m \longrightarrow \mathbb{Z}_q^n$ defined by $H_A(z) = Az \pmod q$.
- **Exercise**: Prove that H_A is collision resistant provided that Ring-SIS is hard.
- Then H_A is a **collision-resistant function** *on average*, provided that SVP_γ for anti-cyclic lattices is hard to solve i*n the worst case*.

Ring-SIS versus SIS

SIS

1. The $n \times m$ matrix A requires storage for mn $\mathbb{Z}_q$ elements.

2. Computing $Ar \pmod q$ takes time $O(mn)$.

Ring-SIS

1. The $n \times m$ matrix $A = [\overline{\mathrm{circ}}(a_1) \,|\, \cdots \,|\, \overline{\mathrm{circ}}(a_\ell)]$ can be derived from $\ell n = m$ $\mathbb{Z}_q$ elements.

2. $Ar \pmod q$ can be computed in time $O(m \log n)$ using the Number-Theoretic Transform (when n divides $q - 1$).

Security

- No attacks (either theoretical or practical) are known on Ring-SIS that are faster than the fastest attacks known on SIS.
- In other words, no attacks are known on Ring-SIS that exploit the structure in the matrix A.

Ring-LWE

- Lyubashevsky-Peikert-Rosen (2010)
- Let S_B denote the polynomials in $R_q = \mathbb{Z}_q[x]/(x^n+1)$ whose coefficients (when reduced mods q) are in $[-B, B]$.
- **Ring-LWE(n, k, q, B):** Let $s \in_R R_q$ and $e_1, \ldots, e_k \in_R S_B$ where $B \ll q/2$.
  Let $a_1, \ldots, a_k \in_R R_q$ and $b_i = a_i s + e_i \in R_q$ for $i = 1, \ldots, k$.
  Given the a_i and b_i, determine s.
- Equivalently, solve the following noisy linear systems of equations for $s \in \mathbb{Z}_q^n$ (and $e \in [-B, B]^{kn}$):

$$As + e = b \quad \begin{bmatrix} \overline{\mathrm{circ}}(a_1) \\ \vdots \\ \overline{\mathrm{circ}}(a_k) \end{bmatrix}_{kn \times n} \begin{bmatrix} s \end{bmatrix}_{n \times 1} + \begin{bmatrix} e_1 \\ \vdots \\ e_k \end{bmatrix}_{kn \times 1} = \begin{bmatrix} b_1 \\ \vdots \\ b_k \end{bmatrix}_{kn \times 1} \pmod q$$

- So, Ring-LWE is a special case of LWE where the matrix A is *structured*.

Example: Ring-LWE (1)

- Let $q = 17$, $n = 4$, $f(x) = x^4 + 1$,
  $R_q = \mathbb{Z}_{17}[x]/(x^4+1)$, $k = 3$, $B = 3$.
- **Ring-LWE instance**:
  Given $a_1(x) = 10 + 16x^2$,
  $a_2(x) = 7 + 10x + 3x^2 + 16x^3$,
  $a_3(x) = 9 + 12x + 16x^2 + 14x^3$,
  $b_1(x) = 16 + 9x + 6x^2 + 4x^3$,
  $b_2(x) = 2 + 16x + 12x^2$,
  $b_3(x) = 10 + 15x + 7x^2$,
  find $s \in R_q$ such that
  $b_i - a_i s = e_i \in S_3$ for $i = 1,2,3$.
- $A = \begin{bmatrix} 10 & 0 & 1 & 0 \\ 0 & 10 & 0 & 1 \\ 16 & 0 & 10 & 0 \\ 0 & 16 & 0 & 10 \\ \hline 7 & 1 & 14 & 7 \\ 10 & 7 & 1 & 14 \\ 3 & 10 & 7 & 1 \\ 16 & 3 & 10 & 7 \\ \hline 9 & 3 & 1 & 5 \\ 12 & 9 & 3 & 1 \\ 16 & 12 & 9 & 3 \\ 14 & 16 & 12 & 9 \end{bmatrix}$ $\quad b = \begin{bmatrix} 16 \\ 9 \\ 6 \\ 4 \\ \hline 2 \\ 16 \\ 12 \\ 0 \\ \hline 10 \\ 15 \\ 7 \\ 0 \end{bmatrix}$.

Example: Ring-LWE (2)

- Solve $As + e = b \pmod{17}$, where $s \in \mathbb{Z}_{17}^4$ and $e \in [-3,3]^{12}$.
- Two solutions:
  - $s = [5, 15, 15, 12]^T$, $e = [2, 0, -3, 1, -2, 1, 2, -2, -2, 1, 1, -3]^T$.
  - $s = [14, 15, 14, 12]^T$, $e = [-2, 0, -1, 1, 0, -3, -1, 0, 3, -2, 2, 2]^T$.
- The first solution in polynomial form is:
  - $s(x) = 5 + 15x + 15x^2 + 12x^3$, $e_1(x) = 2 - 3x^2 + x^3$,
    $e_2(x) = -2 + x + 2x^2 - 2x^3$, $e_3(x) = -2 + x + x^2 - 3x^3$.
  - **Check**: $As + e = b \pmod q$ and $a_i(x)s(x) + e_i(x) = b_i(x)$ in R_q for $i = 1,2,3$.

Ring-LWE security

- Lyubashevsky-Peikert-Rosen proved that solving Ring-LWE on *average* is at least as hard as *quantumly* solving SIVP_γ for anti-cyclic lattices *in the worst case*.
- No attacks (either theoretical or practical) are known on Ring-LWE that are faster than the fastest attacks known on LWE.
- Ring-LWE has the same advantages over LWE as Ring-SIS has over SIS.

$$As + e = b \quad \begin{bmatrix} \overline{\mathrm{circ}}(a_1) \\ \vdots \\ \overline{\mathrm{circ}}(a_k) \end{bmatrix}_{kn \times n} \begin{bmatrix} s \end{bmatrix}_{n \times 1} + \begin{bmatrix} e_1 \\ \vdots \\ e_k \end{bmatrix}_{kn \times 1} = \begin{bmatrix} b_1 \\ \vdots \\ b_k \end{bmatrix}_{kn \times 1} \pmod q$$

Ring-LWE-based public-key encryption

Key generation: Alice does:

1. Select $s \in_R S_B$. [Note: short-secret Ring-LWE]
2. Select $a \in_R R_q$ and $e \in_R S_B$.
3. Compute $b = as + e \in R_q$.
4. Alice's **public key** is (a, b); her **private key** is s.

Encryption: To encrypt a message $m \in \{0,1\}^n$ for Alice, Bob does:

1. Obtain an authentic copy of Alice's encryption key (a, b).
2. Select $r, z, z' \in_R S_B$.
3. Compute $c_1 = ar + z$ and $c_2 = br + z' + \lceil q/2 \rfloor m$.
4. Output $c = (c_1, c_2) \in R_q \times R_q$.

Decryption: To decrypt $c = (c_1, c_2)$, Alice does:

1. Output $\text{Round}_q(c_2 - sc_1)$.

Security: Indistinguishable against chosen-plaintext attacks assuming that Decisional short-secret Ring-LWE is hard.

THE MATHEMATICS OF LATTICE-BASED CRYPTOGRAPHY

7. Module-SIS and Module-LWE

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Outline

1. The module R_q^k
2. Module-SIS (MSIS)
3. Module-LWE (MLWE)

Modules

- **Main idea** in Module-SIS (MSIS)
  Replace polynomials $a_1, a_2, \ldots, a_\ell$ in Ring-SIS by vectors of polynomials in R_q^k.
- **Main idea** in Module-LWE (MLWE)
  Replace polynomials $a_1, a_2, \ldots, a_k$ in Ring-LWE by vectors of polynomials in R_q^ℓ.
- The MSIS and MLWE lattices are less structured than their Ring-SIS and Ring-LWE counterparts.
- Recall: $R = \mathbb{Z}[x]/(x^n + 1)$ and $R_q = \mathbb{Z}_q[x]/(x^n + 1)$, where $n = 2^w$.
- We will work with **modules** R_q^k for MSIS (and R_q^ℓ for MLWE).
  - The module R_q^k is comprised of the length-k vectors of polynomials in R_q.
  - Such vectors can be added and subtracted component-wise, so the result is also a vector in R_q^k.
  - The **inner product** (multiplication) of two vectors in R_q^k results in a polynomial in R_q.
  - The **size** of $a = (a_1, a_2, \ldots, a_k) \in R_q^k$ is $\|a\|_\infty = \max_i \|a_i\|_\infty$.
- See V1b of my "Kyber and Dilithium" course for examples.

Module-SIS (1)

- **MSIS**(n, k, ℓ, q, B):
  Given $a_1, a_2, \ldots, a_\ell \in_R R_q^k$ (where $\ell > k$), find $z_1, z_2, \ldots, z_\ell \in R_q$ such that $a_1 z_1 + a_2 z_2 + \cdots + a_\ell z_\ell = 0$ where $\|z_i\|_\infty \le B$ and not all z_i are 0.
- **Note**: Each a_i is now a vector of polynomials: $a_i = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{ik}]^T$.
- So, Module-SIS asks for a "small" nonzero solution to the polynomial-matrix equation:

$$\begin{bmatrix} a_{11} & a_{21} & \cdots & a_{\ell 1} \\ a_{12} & a_{22} & \cdots & a_{\ell 2} \\ \vdots & \vdots & & \vdots \\ a_{1k} & a_{2k} & \cdots & a_{\ell k} \end{bmatrix}_{k \times \ell} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_\ell \end{bmatrix}_{\ell \times 1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{k \times 1}.$$

- **Note**: If $(z_1, z_2, \ldots, z_\ell)$ is a solution then so is $(xz_1, xz_2, \ldots, xz_\ell)$.

Module-SIS (2)

- Equivalent formulation of **MSIS**(n, k, ℓ, q, B):
  Given $a_1, a_2, \ldots, a_\ell \in_R R_q^k$, find nonzero $z \in [-B, B]^m$ (where $m = \ell n$) such that $Az = 0 \pmod q$, where

$$A = \begin{bmatrix} \overline{\text{circ}}(a_{11}) & \overline{\text{circ}}(a_{21}) & \cdots & \overline{\text{circ}}(a_{\ell 1}) \\ \overline{\text{circ}}(a_{12}) & \overline{\text{circ}}(a_{22}) & \cdots & \overline{\text{circ}}(a_{\ell 2}) \\ \vdots & \vdots & & \vdots \\ \overline{\text{circ}}(a_{1k}) & \overline{\text{circ}}(a_{2k}) & \cdots & \overline{\text{circ}}(a_{\ell k}) \end{bmatrix}_{kn \times \ell n}.$$

- So, MSIS is a special case of SIS where the matrix A is *structured*.

Example: MSIS (1)

- Let $q = 67$, $n = 4$, $f(x) = x^4 + 1$, $R_q = \mathbb{Z}_{67}[x]/(x^4 + 1)$, $k = 2$, $\ell = 3$, $B = 10$.
- Let $a_1 = [a_{11}, a_{12}]^T = [32 + 66x^2 + 33x^3,\ 30 + 64x + 31x^2 + 65x^3]^T \in R_q^2$,
  $a_2 = [a_{21}, a_{22}]^T = [42 + 44x + 20x^2 + 65x^3,\ 63 + 41x + 19x^2 + 64x^3]^T \in R_q^2$,
  $a_3 = [a_{31}, a_{32}]^T = [2 + 60x + 33x^2 + 42x^3,\ 26 + 9x + 57x^2 + 7x^3]^T \in R_q^2$.
- **MSIS instance:** Find $z_1, z_2, z_3 \in R_q$, not all 0, with $a_{11}z_1 + a_{21}z_2 + a_{31}z_3 = 0 \pmod q$, $a_{12}z_1 + a_{22}z_2 + a_{32}z_3 = 0 \pmod q$, and $\|z_i\|_\infty \le 10$.
- We have

$$A = \left[\begin{array}{cccc|cccc|cccc} 32 & 34 & 1 & 0 & 42 & 2 & 47 & 23 & 2 & 25 & 34 & 7 \\ 0 & 32 & 34 & 1 & 44 & 42 & 2 & 47 & 60 & 2 & 25 & 34 \\ 66 & 0 & 32 & 34 & 20 & 44 & 42 & 2 & 33 & 60 & 2 & 25 \\ 33 & 66 & 0 & 32 & 65 & 20 & 44 & 42 & 42 & 33 & 60 & 2 \\ \hline 30 & 2 & 36 & 3 & 63 & 3 & 48 & 26 & 26 & 60 & 10 & 58 \\ 64 & 30 & 2 & 36 & 41 & 63 & 3 & 48 & 9 & 26 & 60 & 10 \\ 31 & 64 & 30 & 2 & 19 & 41 & 63 & 3 & 57 & 9 & 26 & 60 \\ 65 & 31 & 64 & 30 & 64 & 19 & 41 & 63 & 7 & 57 & 9 & 26 \end{array}\right]_{8 \times 12}.$$

Example: MSIS (2)

- Gaussian elimination (mod q) on A yields the following matrix in reduced form:

$$A' = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 17 & 27 & 21 & 28 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 39 & 17 & 27 & 21 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 46 & 39 & 17 & 27 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 40 & 46 & 39 & 17 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 46 & 29 & 44 & 53 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 14 & 46 & 29 & 44 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 23 & 14 & 46 & 29 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 38 & 23 & 14 & 46 \end{bmatrix}.$$

- The set of all solutions $r = (r_1, r_2, \ldots, r_{12}) \in \mathbb{Z}_{59}^{12}$ to $A'r = 0 \pmod q$ is:

$$\begin{aligned} r_1 &= 50r_9 + 40r_{10} + 46r_{11} + 39r_{12} & r_2 &= 28r_9 + 50r_{10} + 40r_{11} + 46r_{12} \\ r_3 &= 21r_9 + 28r_{10} + 50r_{11} + 40r_{12} & r_4 &= 27r_9 + 21r_{10} + 28r_{11} + 50r_{12} \\ r_5 &= 21r_9 + 38r_{10} + 23r_{11} + 14r_{12} & r_6 &= 53r_9 + 21r_{10} + 38r_{11} + 23r_{12} \\ r_7 &= 44r_9 + 53r_{10} + 21r_{11} + 38r_{12} & r_8 &= 29r_9 + 44r_{10} + 53r_{11} + 21r_{12}. \end{aligned}$$

Example: MSIS (3)

- The total number of solutions to $A'r = 0 \pmod q$ is $q^4 = 20{,}151{,}121$.
  - Of these, the number of solutions r that are nonzero and in $[-10, 10]^{12}$ is 8.
- The MSIS solution (up to multiplication by $\pm 1,\ \pm x,\ \pm x^2,\ \pm x^3$) is:
  $r = (6, -8, 8, 0, 2, 10, -6, 3, -9, 6, 3, 2)$
- The solution in polynomial form is:
  $z_1(x) = 6 - 8x + 8x^2,\ \ z_2(x) = 2 + 10x - 6x^2 + 3x^3,\ \ z_3(x) = -9 + 6x + 3x^2 + 2x^3.$
- **Check**:
  (i) $Ar = 0 \pmod q$,
  (ii) $a_{11}(x)z_1(x) + a_{21}(x)z_2(x) + a_{31}(x)z_3(x) = 0$ in R_q, and
  (iii) $a_{12}(x)z_1(x) + a_{22}(x)z_2(x) + a_{32}(x)z_3(x) = 0$ in R_q.

Module-SIS notes

- Langlois and Stehlé (2015) introduced MSIS and proved that solving MSIS on *average* is at least as hard as solving SIVP_γ for module lattices *in the worst case.*
- Setting $k = 1$ gives an instance of Ring-SIS.
- Setting $n = 1$ (replacing R_q by $\mathbb{Z}_q$) gives an instance of SIS.
- So, MSIS "interpolates" between SIS and Ring-SIS.

$$A = \begin{bmatrix} \overline{\mathrm{circ}}(a_{11}) & \overline{\mathrm{circ}}(a_{21}) & \cdots & \overline{\mathrm{circ}}(a_{\ell 1}) \\ \overline{\mathrm{circ}}(a_{12}) & \overline{\mathrm{circ}}(a_{22}) & \cdots & \overline{\mathrm{circ}}(a_{\ell 2}) \\ \vdots & \vdots & & \vdots \\ \overline{\mathrm{circ}}(a_{1k}) & \overline{\mathrm{circ}}(a_{2k}) & \cdots & \overline{\mathrm{circ}}(a_{\ell k}) \end{bmatrix}_{kn \times \ell n}$$

- A primary advantage of MSIS over Ring-SIS is that parameters q and n can be fixed for MSIS, and then k can be varied for different security levels.
- For example, Dilithium fixes $q = 8380417,\ n = 256$, and $(k, \ell) \in \{(4,4), (6,5), (8,7)\}$, where now the underlying matrix of polynomials is $[A \mid I_k]_{k \times \ell}$ where $A \in_R R_q^{k \times \ell}$.
  So, Dilithium is "closer" to Ring-SIS than to SIS.
- Since $2n = 512$ divides $q - 1$, the Number-Theoretic Transform can be used for fast polynomial multiplication in $R_q = \mathbb{Z}_q[x]/(x^{256} + 1)$.

Module-LWE

- **MLWE(n, k, ℓ, q, B)**:
  Let $s \in_R R_q^\ell$ and $e \in_R S_B^k$ where $k > \ell$ and $B \ll q/2$.
  Let $a_1, a_2, \ldots, a_k \in_R R_q^\ell$ and $b_i = a_i^T s + e_i \in R_q$ for $i = 1, \ldots, k$.
  Given the a_i and b_i, determine s.
- Note that each a_i is now a vector of polynomials: $a_i = [a_{i1} \quad a_{i2} \quad \cdots \quad a_{i\ell}]^T$.
- So, Module-LWE asks for a solution $s \in R_q^\ell$, $e \in S_B^k$ to the polynomial-matrix equation:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1\ell} \\ a_{21} & a_{22} & \cdots & a_{2\ell} \\ \vdots & \vdots & & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{k\ell} \end{bmatrix}_{k \times \ell} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_\ell \end{bmatrix}_{\ell \times 1} + \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_k \end{bmatrix}_{k \times 1} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \end{bmatrix}_{k \times 1}.$$

Module-LWE (2)

- Equivalent formulation of **MLWE(n, k, ℓ, q, B)**:
  Let $s \in_R R_q^\ell$ and $e \in_R S_B^k$ where $k > \ell$ and $B \ll q/2$.
  Let $a_1, a_2, \ldots, a_k \in_R R_q^\ell$ and $b_i = a_i^T s + e_i \in R_q$ for $i = 1, \ldots, k$.

  Given the a_i and b_i, find $s \in \mathbb{Z}_q^{\ell n}$ and $e \in [-B, B]^{kn}$ such that $As + e = b \pmod q$, where

$$A = \begin{bmatrix} \overline{\mathrm{circ}}(a_{11}) & \overline{\mathrm{circ}}(a_{21}) & \cdots & \overline{\mathrm{circ}}(a_{\ell 1}) \\ \overline{\mathrm{circ}}(a_{12}) & \overline{\mathrm{circ}}(a_{22}) & \cdots & \overline{\mathrm{circ}}(a_{\ell 2}) \\ \vdots & \vdots & & \vdots \\ \overline{\mathrm{circ}}(a_{1k}) & \overline{\mathrm{circ}}(a_{2k}) & \cdots & \overline{\mathrm{circ}}(a_{\ell k}) \end{bmatrix}_{kn \times \ell n} \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \end{bmatrix}_{kn \times 1}.$$

- So, MLWE is a special case of LWE where the matrix A is *structured*.

Example: Module-LWE (1)

- Let $q = 37,\ n = 4,\ f(x) = x^4 + 1,\ R_q = \mathbb{Z}_{37}[x]/(x^4+1),\ k = 3,\ \ell = 2,\ B = 1.$
- **Module-LWE instance**: Given

$$a_1 = \begin{bmatrix} a_{11}(x) \\ a_{21}(x) \\ a_{31}(x) \end{bmatrix} = \begin{bmatrix} 21 + 5x^2 + 19x^3 \\ 4 + 14x + 20x^2 + 19x^3 \\ 13 + 7x + 6x^2 + 8x^3 \end{bmatrix}, \quad a_2 = \begin{bmatrix} a_{12}(x) \\ a_{22}(x) \\ a_{32}(x) \end{bmatrix} = \begin{bmatrix} 1 + 23x + 9x^2 + 8x^3 \\ 24 + 23x + 22x^2 + 21x^3 \\ 34 + 33x + 32x^2 + 31x^3 \end{bmatrix},$$

$$\text{and } b = \begin{bmatrix} b_1(x) \\ b_2(x) \\ b_3(x) \end{bmatrix} = \begin{bmatrix} 19 + 32x + 7x^2 + 20x^3 \\ 32 + 19x^2 + 8x^3 \\ 29 + 15x + 11x^2 + 14x^3 \end{bmatrix},$$

find $s = \begin{bmatrix} s_1(x) \\ s_2(x) \end{bmatrix} \in R_q^2$ such that $b_i(x) - a_{i1}(x)s_1(x) - a_{i2}(x)s_2(x) = e_i(x) \in S_1$ for $i = 1,2,3$.

Example: Module-LWE (2)

- Solve $As + e = b \pmod{37}$ for $s \in \mathbb{Z}_{37}^8$ and $e \in [-1,1]^{12}$, where

$$A = \left[\begin{array}{cccc|cccc} 21 & 18 & 32 & 0 & 1 & 29 & 28 & 14 \\ 0 & 21 & 18 & 32 & 23 & 1 & 29 & 28 \\ 5 & 0 & 21 & 18 & 9 & 23 & 1 & 29 \\ 19 & 5 & 0 & 21 & 8 & 9 & 23 & 1 \\ \hline 4 & 18 & 17 & 23 & 24 & 16 & 15 & 14 \\ 14 & 4 & 18 & 17 & 23 & 24 & 16 & 15 \\ 20 & 14 & 4 & 18 & 22 & 23 & 24 & 16 \\ 19 & 20 & 14 & 4 & 21 & 22 & 23 & 24 \\ \hline 13 & 29 & 31 & 30 & 34 & 6 & 5 & 4 \\ 7 & 13 & 29 & 31 & 33 & 34 & 6 & 5 \\ 6 & 7 & 13 & 29 & 32 & 33 & 34 & 6 \\ 8 & 6 & 7 & 13 & 31 & 32 & 33 & 34 \end{array}\right]_{12\times 8} \quad \text{and } b = \left[\begin{array}{c} 19 \\ 32 \\ 7 \\ 20 \\ \hline 32 \\ 0 \\ 19 \\ 8 \\ \hline 29 \\ 15 \\ 11 \\ 14 \end{array}\right]_{12\times 1}.$$

Example: Module-LWE (3)

- Solve $As + e = b \pmod{37}$, where $s \in \mathbb{Z}_{37}^8$ and $e \in [-1,1]^{12}$.
- There are two solutions (s, e):
  - $s = [31, 32, 33, 2, 17, 35, 13, 32],\ e = [-1, 0, -1, 1, 0, -1, -1, 0, 1, 0, 0, 1]^T.$
  - $s = [2, 29, 9, 22, 32, 12, 27, 18],\ e = [-1, 0, 1, 0, 1, -1, 0, -1, 1, -1, -1, 0]^T.$
- The first solution in polynomial form is:
  $s_1(x) = 31 + 32x + 33x^2 + 2x^3,\ s_2(x) = 17 + 35x + 13x^2 + 32x^3,$
  $e_1(x) = -1 - x^2 + x^3,\ e_2(x) = -x - x^2,\ e_3(x) = 1 + x^3.$
  - **Check**: $As + e = b \pmod{37}$
    and $a_{i1}(x)s_1(x) + a_{i2}s_2(x) + e_i(x) = b_i(x)$ in R_q for $i = 1,2,3$.

Module-LWE notes

- Module-LWE was introduced by Brakerski, Gentry and Vaikuntanathan (2011).
- <u>Check</u>: Setting $\ell = 1$ gives an instance of Ring-LWE.
- Setting $n = 1$ (replacing R_q by $\mathbb{Z}_q$) gives an instance of LWE.
- So, MLWE "interpolates" between LWE and Ring-LWE.

$$A = \begin{bmatrix} \overline{\text{circ}}(a_{11}) & \overline{\text{circ}}(a_{21}) & \cdots & \overline{\text{circ}}(a_{\ell 1}) \\ \overline{\text{circ}}(a_{12}) & \overline{\text{circ}}(a_{22}) & \cdots & \overline{\text{circ}}(a_{\ell 2}) \\ \vdots & \vdots & & \vdots \\ \overline{\text{circ}}(a_{1k}) & \overline{\text{circ}}(a_{2k}) & \cdots & \overline{\text{circ}}(a_{\ell k}) \end{bmatrix}_{kn\times \ell n}$$

- Langlois and Stehlé (2015) proved that solving MLWE on *average* is at least as hard as *quantumly* solving SIVP_γ for module lattices *in the worst case*.
- However, as with Regev's worst-case to average-case reduction for LWE, the reduction is *highly non-tight* (and also a quantum reduction).
  - For a concrete analysis of the Langlois-Stehlé reduction for MLWE (and also the Lyubashevsky-Peikert-Regev reduction for Ring-LWE) see:
    "Concrete analysis of approximate Ideal-SIVP to Decision Ring-LWE reduction"
    by Koblitz, Samajder, Sarkar and Singha, <u>https://eprint.iacr.org/2022/275</u>.

MLWE versus Ring-LWE

- A primary advantage of MLWE over Ring-LWE is that parameters q and n can be fixed for MLWE, and then ℓ can be changed for different security levels.
- For example, Dilithium fixes $q = 8380417$ and $n = 256$, and $(k, \ell) \in \{(4,4), (6,5), (8,7)\}$.
  - So, one can optimize arithmetic in the polynomial ring $R_q = \mathbb{Z}_{8380417}[x]/(x^{256} + 1)$.
  - Dilithium is "closer" to Ring-LWE than to LWE.

Kyber-PKE: Key generation

Key generation: Alice does:

1. Select $s \in_R S_{\eta_1}^k$.
2. Select $A \in_R R_q^{k \times k}$ and $e \in_R S_{\eta_1}^k$.
3. Compute $b = As + e$.
4. Alice's **public key** is (A, b); her **private key** is s.

- $q = 3329, \quad n = 256$.
- $R_q = \mathbb{Z}_{3329}[x]/(x^{256} + 1)$.
- $k \in \{2,3,4\}$.
- $(\eta_1, \eta_2) \in \{(3,2), (2,2), (2,2)\}$.

- Computing s from (A, b) is an instance of **ss-MLWE**.
- Determining any information about s from (A, b) is an instance of **ss-DMLWE**.

Kyber-PKE: Encryption and Decryption

Encryption: To encrypt a message $m \in \{0,1\}^{256}$ for Alice, Bob does:

1. Obtain an authentic copy of Alice's encryption key (A, b).
2. Select $r \in_R S_{\eta_1}^k$, $z \in_R S_{\eta_2}^k$ and $z' \in_R S_{\eta_2}$.
3. Compute $c_1 = A^T r + z$ and $c_2 = b^T r + z' + \lceil q/2 \rfloor m$.
4. Output $c = (c_1, c_2)$.

<u>**Note**</u>: $c \in R_q^k \times R_q$.

Decryption: To decrypt $c = (c_1, c_2)$, Alice does:

1. Compute $m = \mathrm{Round}_q(c_2 - s^T c_1)$.

Security:
Kyber-PKE is indistinguishable against chosen-plaintext attack assuming the hardness of short-secret Decisional Module-LWE.

Security

- No attacks (either theoretical or practical) are known on Module-SIS or Module-LWE that are any faster than the fastest attacks known on SIS and LWE.
- In other words, no attacks are known on Module-SIS or Module-LWE that exploit the structure in the matrix A.
- The fastest attacks known on SIS and LWE (see Lecture 5) are used to select MSIS and MLWE parameters in order to attain a desired security level.
  - See the "Bochum challenges": https://bochum-challeng.es